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Distributed cortical learning through LEC-mediated γ-synchrony

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Despite the well-established theoretical and experimental foundations of dopamine-driven reinforcement learning, how the reward prediction error (RPE) teaching signal modifies specific cortical memory networks remains unclear. Neural oscillations are crucial for the temporal binding of activities across distributed cell ensembles. Here, we identify that mouse lateral entorhinal cortex layer 5 (LEC^5) mediated intercortical γ-synchrony is a neural correlate of the RPE signal derived from dopamine neurons in the ventral tegmental area (VTA^{DA}). The VTA^{DA}-LEC^5 circuit-based intercortical γ-synchrony facilitates both learning and memory retention, and at the single-cell level, entrains the activity of cortical latent Engrams. Human brain recordings also validate a role of γ-synchrony in the processing of prediction errors. These findings indicate that LEC^5-mediated intercortical γ-synchrony functions as a reinforcement learning signal that facilitates the establishment of cortical memory networks.

Learning helps animals adapt to environmental changes much more rapidly than evolution does. Internalizing the acquired knowledge into lasting modifications within the neural networks facilitates prompt and energy-efficient responses to analogous situations rooted in previous experience. According to the reward prediction error hypothesis of midbrain dopamine neurons within the temporal-difference (TD) learning model, the discrepancy between expectations of all future rewards and any information that leads to a revision of these expectations serves as the driving force of reinforcement learning[1–4]. On the other hand, event-related Engrams, which are characterized as the cellular substrates responsible for the encoding and retrieval of long-term memory[5,6], are temporally localized within a specific time window and are sparsely distributed throughout the brain[7–9]. This raises a fundamental question: how does the RPE teaching signal coordinate this sparse yet extensive Engram network in the brain?

Widespread neural oscillations across a spectrum of frequencies provide various temporal windows for instantaneous communications among segregated brain areas, reflecting diverse modes of functional connectivity in neural networks during information processing[10–13]. Gamma band oscillations (γ-oscillations) were implicated in associative learning and memory storage processes in both mice and humans[14–16]. Furthermore, our prior research demonstrated that artificially inducing the synchronization of cortical γ-oscillations (intercortical γ-synchrony) via stimulation of the LEC^5 reinstated contextual memory formation and retrieval in mice with hippocampal lesions[17]. Nonetheless, the precise biological function of intercortical γ-synchrony in learning and memory remains unclear.

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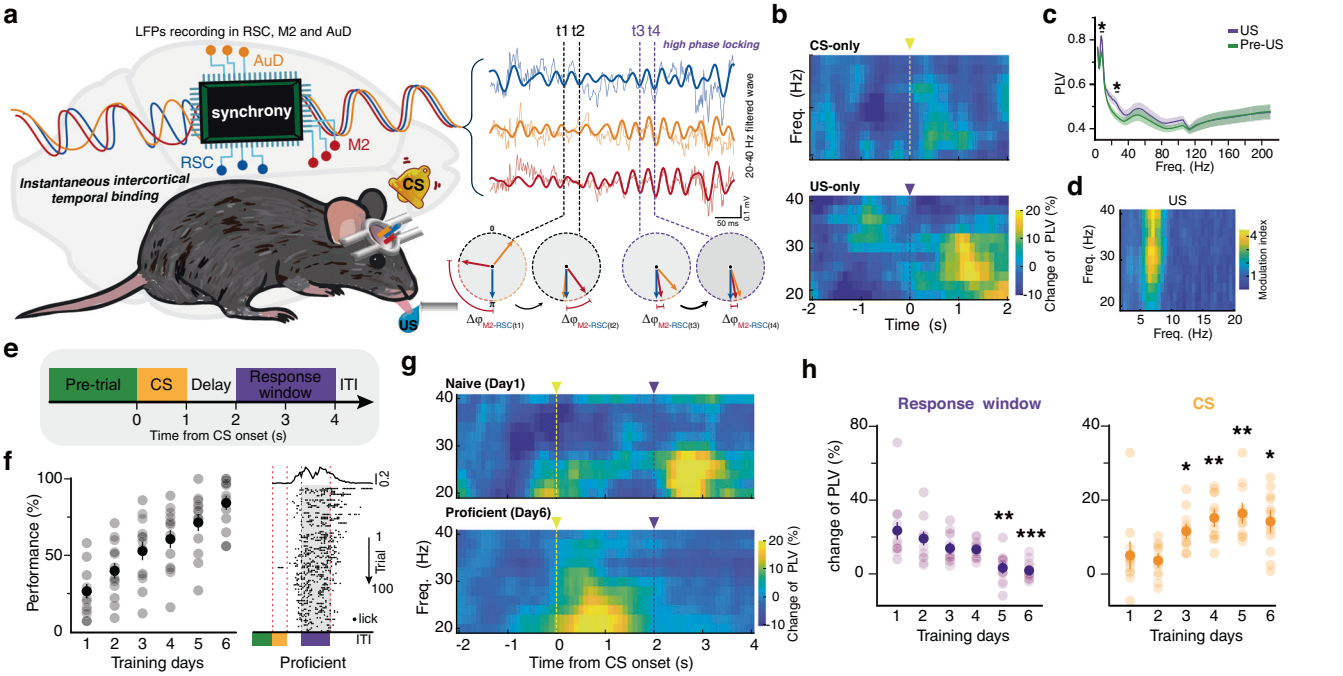


Fig. 1 | Intercortical γ-synchrony dynamics reflect the reward prediction error signal. a Left, schematic depiction of multi-region (RSC, AuD, M2) LFPs recording during a CS-US associative learning task. The mouse illustration is adapted from https://www.magnific.com/free-vector/creative-hand-drawn-mice-collection_1587915.htm. Right, representative γ-band (20–40 Hz, dark lines) phase extraction from LFP raw traces (tint lines). Phase locking value (PLV) measures the instantaneous phase difference (Δφ) between regional waves. Using $\Delta\varphi_{M2\text{-}RSC}$ as an example, t1-t2 and t3-t4 were defined by consecutive RSC γ-troughs in two time segments. The more stable Δφ at t3-t4 versus t1-t2 points to increased RSC-M2 synchrony. **b** Averaged γ-synchrony spectrogram for CS-only trials (top) and US-only trials (bottom). $n = 39$ cortical pairs from 13 mice. The triangle and dashed line indicate the onset times for CS (yellow) or US (purple) delivery. **c** Comparison of the averaged overall synchrony between the US and pre-US time segments. $P_{8Hz\ band} = 0.014$, $P_{20\text{-}30Hz\ band} = 0.036$. $n = 13$ mice. Shading: S.E.M. **d** Normalized phase-power modulation comodulogram within the first 2 s post-US. $n = 39$ cortical pairs from 13 mice. **e** Trial structure of CS-US associative training across day1 to day6. **f** Left, performance of all 13 mice. Each dot represents one mouse. The dark dot indicates the average value. Error bar shows S.E.M. Right, representative lick behavior of a proficient mouse (day6). Licking density peaked within the response window over learning. **g** Spectrogram of averaged γ-synchrony for naive (top, day1) and proficient trials (bottom, day6). The triangle and dotted line indicate the onset time for CS (yellow) and the response window (purple). **h** Evolution of intercortical γ-synchrony within the response window (left) and during a 1 s CS-lasting period (right), respectively. For the response window, $P_{day5\ vs.\ day1} = 0.004$, $P_{day6\ vs.\ day1} = 0.001$. For the CS period, $P_{day3\ vs.\ day1} = 0.045$, $P_{day4\ vs.\ day1} = 0.008$, $P_{day5\ vs.\ day1} = 0.003$, $P_{day6\ vs.\ day1} = 0.047$. $n = 13$ mice. Each dot represents one mouse. The dark dot indicates the average value. Error bar shows S.E.M. $^*P < 0.05$, $^{***}P < 0.001$. P values were calculated using two-way ANOVA Sidak method (**c**), and one-way ANOVA multiple comparisons Dunnett method (**h**). Source data are provided as a Source data file.

In this study, we explored the correlation between the intercortical γ-synchrony and the dopamine-driven RPE signal. We combined fiber photometry, electrophysiology recordings, and two-photon (2P) imaging techniques to uncover the underlying neural circuit mechanisms. Moreover, we verified the causal relationship between circuit-based intercortical γ-synchrony and animal behaviors. At the single-cell level, to understand how intercortical γ-synchrony entrains the activity of latent cortical Engrams, we recorded calcium activity in retrosplenial cortex (RSC) neurons and local field potentials (LFPs) in both RSC and auditory cortex (AuD) simultaneously through a combination of 2P microscopy and ultraflexible electrode arrays. Finally, we conducted electrocorticography (ECoG) recordings in human brains during a real-phrases or pseudo-phrases reading task to examine the cross-species generalizability of intercortical γ-synchrony in the processing of prediction error.

Results

Intercortical γ-synchrony reflects the RPE signal processing

In Pavlovian conditioning, through trial and error, anticipation of the unconditioned stimulus (US) following the neutral conditioned stimulus (CS) is indicative of adaptive changes made by the nervous system. We designed a Pavlovian tone-reward (CS-US) association learning paradigm with specific trial structures. LFPs from three task-related cortical regions, including RSC, AuD, and the secondary motor cortex (M2), were recorded during the behavior experiment (Fig. 1a). The strength of coupled synchrony between pairwise oscillations can be quantified using the phase-locking value (PLV)[17]. Elevated PLV indicates a stable pairwise phase difference, suggesting increased coordinated activity between two brain regions.

Initially, we exposed water-restricted mice ($n = 13$) to ten CS-only trials and ten US-only trials, respectively. No significant pairwise PLV change was observed during the CS-only trials. Conversely, after receiving a reward in the US-only trials, there were significant increases (13.50 ± 1.71%) in intercortical γ-synchrony (20–40 Hz, averaged across three pairs: RSC-AuD, RSC-M2, and AuD-M2) compared to the baseline (Fig. 1b). Additionally, the levels of theta band (6–10 Hz) synchrony (Fig. 1c and Supplementary Fig. 1d) and theta-gamma cross-frequency modulation index (Fig. 1d and Supplementary Fig. 1e–g) also increased, indicating stronger cross-frequency coupling. The conjecture that licking behavior contributed to the increase in intercortical γ-synchrony is not supported by a comparison of the PLV between periods when mice engaged in licking water and periods when they did not (Supplementary Fig. 1a). There was no significant difference in the extent to which each pair contributed to the increase in γ-synchrony (Supplementary Fig. 1b). Because the power of γ-oscillations was not significantly increased during the task (Supplementary Fig. 1c) and the main change was in phase synchronization, we decided not to use the coherence measurement. Our prior work also showed that intercortical γ-synchrony is unaffected by volume conduction[17].

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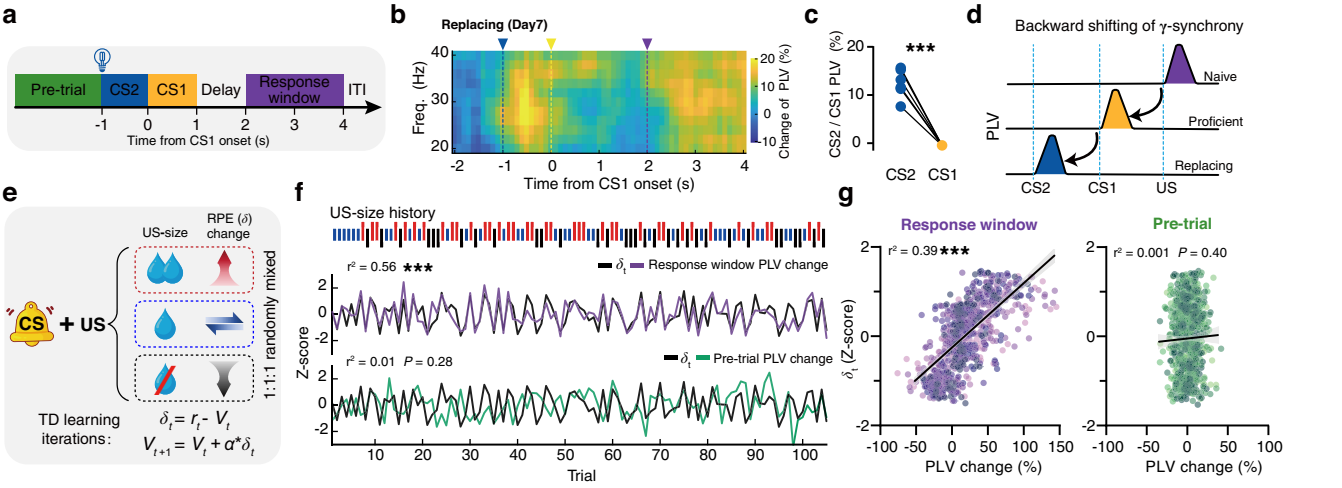


Fig. 2 | Intercortical γ-synchrony dynamics conform to the TD learning algorithm. a Trial structure for CS replacing experiment on day7. **b** Spectrogram of averaged γ-synchrony for the CS replacing experiment (day7, $n = 6$ mice). **c** Relative changes of averaged intercortical γ-synchrony during the CS2 period, normalized to the CS1 period. $P < 0.0001$. $n = 6$ mice. **d** Diagram illustrating the backward shift of intercortical γ-synchrony accompanied by the RPE signal. **e** Design of the trial-by-trial US size fluctuation experiment on day7. Three trial types, including regular US size (blue), double US size (red), and US omission (black) trials, were randomly mixed in a 1:1:1 ratio. δ: RPE. **f** Top, representative reward history (bars in the same color coding as in e). Middle, a line diagram showing changes in PLV within the response window (purple), normalized to the baseline (averaged PLV within 6 s before CS), and trial-by-trial RPE value changes (black). $P < 0.0001$. Bottom, the changes in PLV within 2 s before CS (green), normalized to baseline, and trial-by-trial RPE value changes (black). $P = 0.28$. **g** Linear regression of trial-by-trial RPE fluctuations with intercortical γ-synchrony changes within the response window ($r^2 = 0.39$, $P < 0.001$, purple) and within the last 2 s pre-trial period ($r^2 = 0.001$, $P = 0.40$, green), $n = 7$ mice. Each dot shows one trial. The gradient color indicates different mice. Shading: 95% confidence bands. ***$P < 0.001$. P values were calculated using two-tailed paired t-test (**c**), and the two-sided Pearson correlation test (**f**, **g**). Source data are provided as a Source data file.

All mice underwent daily 100 CS-US associative trials throughout training from day1 to day6. In each trial, the CS tone persists for 1 s, followed by a 1 s delay, then a 2-s response window. Only licking the water pipe within the response window will trigger the infrared sensor to pump a drop of water as a reward. Each trial is separated by a 5–8 s inter-trial interval (ITI), followed by a 4–6 s lick-withhold period (Fig. 1e). After six days of training, the average behavioral performance reached 84.31 ± 4.60%, attaining a proficient level, with licking movements progressively stabilizing within the response window (Fig. 1f and Supplementary Fig. 1h). A day-by-day analysis of the PLV changes revealed a backward shift in intercortical γ-synchrony from the response window (2 to 4 s) to the CS lasting period (0 to 1 s) as associative training progressed (Fig. 1g). Moreover, intercortical γ-synchrony within the response window gradually declined, while it exhibited an opposite trend during the CS period (Fig. 1h). During the learning task, theta band synchronization increased in both the CS and US phases (Supplementary Fig. 1i). Cross-correlation with the water-licking curves showed that theta band synchronization might be linked to water-licking actions rather than carrying the RPE information (Supplementary Fig. 1j, k).

Based on the TD learning model, it can be inferred that the RPE signal traces back to the earliest cue capable of predicting rewards and terminates there, as no earlier predictive signal exists than this one[4]. To further confirm the relationship between the γ-synchrony backward shift phenomenon and RPE signal encoding, we conducted a CS replacing experiment on day7 with six proficient mice from the earlier CS-US association training. In this experiment, a blue light cue (CS2, −1 to 0 s) was presented ahead of the identical tone cue (CS1) (Fig. 2a). The average behavioral performance exceeded 70% on day7 (Supplementary Fig. 1,1). Once again, the enhancement of γ-synchrony diminished during the CS1 period (devalued) and persisted in shifting backward in time towards CS2 (Fig. 2b–d).

Finally, to assess the sensitivity of intercortical γ-synchrony in reflecting the dynamic trial-by-trial fluctuations in the RPE signal, we adjusted the CS-US contingency by randomly varying the US size in each trial on day7 for the other seven remaining proficient mice (Fig. 2e). The average behavioral performance surpassed 70% on day7 (Supplementary Fig. 1m). According to the TD learning algorithm, rewards that surpass expectations elicit a positive RPE signal, whereas rewards that fall below expectations generate a negative RPE signal. The trial-by-trial relative changes in RPE values following the delivery of different rewards to mice can be calculated iteratively[18]. During the iterative calculations, trials in which animals abstained from water-licking activities throughout the entire duration from CS to ITI were excluded, as such trials signified incomplete engagement with the environment. The relative difference in the average intercortical γ-synchrony between the baseline and response window for trials excluded due to lack of action was not statistically significant (Supplementary Fig. 1n). The initial expectation of reward (VI) for each mouse was set to its hit rate on day6, with a learning rate (α) of 0.5. Linear fitting analysis demonstrated a statistically significant positive correlation between the relative PLV change within the response window (compared to baseline, −6 to 0 s) and the current RPE update, in contrast to the PLV change within the pre-trial period (−2 to 0 s) (Fig. 2f, g).

Intercortical γ-synchrony correlates with the VTA^{DA} activity

We then used fiber photometry and LFP recordings to measure the correlation between the intercortical γ-synchrony and VTA^{DA} population activities during associative learning in DAT-Cre mice (Fig. 3a, b). The training paradigm was consistent with that depicted in Fig. 1e. As training progressed, the enhancement of intercortical γ-synchrony during the CS period and the calcium activities of VTA^{DA} were positively correlated (Fig. 3c, d and Supplementary Fig. 2a). Notably, the time lag between the onset of calcium activities in the VTA^{DA} population during the CS period and the increase in intercortical γ-synchrony on day6 revealed that VTA^{DA} activity preceded the rise in PLV (0.18 ± 0.04 s), indicating that VTA^{DA} is upstream in the RPE information cascade. (Fig. 3e). Furthermore, optogenetic stimulation of the unilateral VTA^{DA} at 20 Hz (approximating the VTA^{DA} phasic firing rate[4]) elicited a consistent augmentation (82.47 ± 9.12%) in ipsilateral intercortical γ-synchrony following a brief delay (0.45 ± 0.05 s, Fig. 3f–i and

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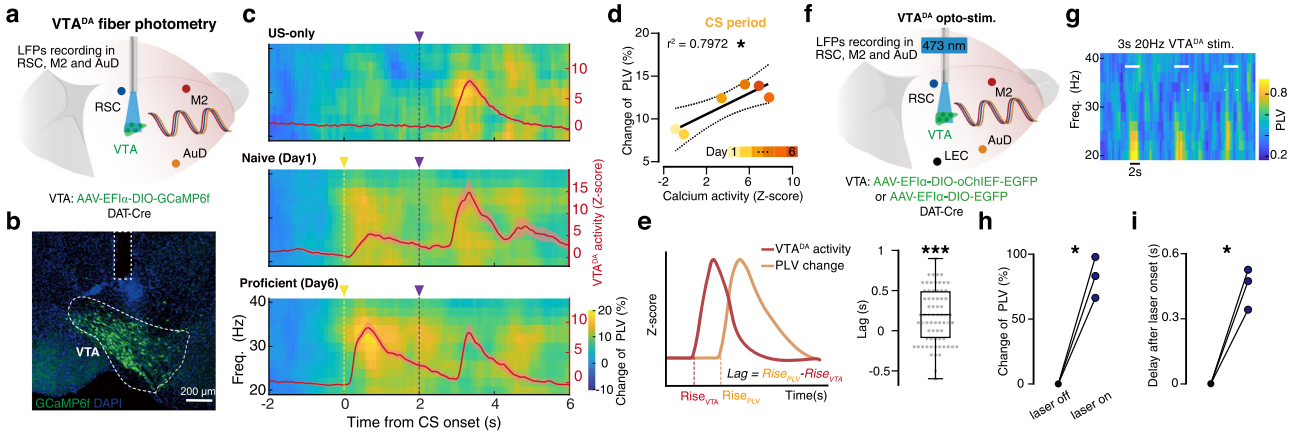


Fig. 3 | Intercortical γ-synchrony correlates with the VTA^{DA} activity. a Viral strategy for expressing GCaMP6f in VTA^{DA} of DAT-Cre mice. Cortical LFPs of ipsilateral RSC, M2, and AuD were recorded simultaneously. **b** Representative Cre-mediated expression of GCaMP6f in VTA^{DA} and optical fiber placement. **c** One mouse example (out of 6 mice) of trial-averaged intercortical γ-synchrony spectrograms with population activities of VTA^{DA} across different stages. Shading: S.E.M. **d** Day-by-day correlation between trial-averaged population activity of VTA^{DA} and the changes in intercortical γ-synchrony within the CS period. $P = 0.017$, $n = 6$ mice. **e** Left, method for calculating the time lag between VTA^{DA} calcium activity (red) and intercortical γ-synchrony (yellow) within the CS period. Right, quantification of time lags on day6, compared to zero, $P < 0.0001$, $n = 70$ trials. Whiskers show min and max, box shows 25th, median and 75th percentile. **f** Viral strategy for expressing oChIEF or EGFP in VTA^{DA} of DAT-Cre mice. An optical fiber was implanted above the VTA. LFPs in LEC, RSC, M2, and AuD of the ipsilateral hemisphere were recorded. **g** Representative spectrogram (out of 3 mice) of intercortical γ-synchrony before and after optogenetic stimulation. White lines indicate laser stimuli. The spectrogram is an average of all cortical pairs. **h** Quantification of PLV changes before and after optogenetics stimulation. $P = 0.012$, $n = 3$ mice. **i** Time lags between the onset of the laser and PLV elevation. $P = 0.015$, $n = 3$ mice. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. P values were calculated using two-sided Pearson correlation test (**d**), and two-tailed paired t-test (**e**, **h**, **i**). Source data are provided as a Source data file.

Supplementary Fig. 2g, h). The observed reduction in time lag to 0.18 s in fiber photometry recordings, compared to 0.45 s associated with optogenetic stimulation, may be attributable to the slower dynamics of the calcium reporters. This alignment of the slow calcium signal with the LFP signals consequently reduces the delay.

Following VTA^{DA} stimulation, a discernible upward trend in γ-synchrony was observed between the LEC and cortices, as well as in LEC γ-oscillation power (Supplementary Fig. 2b–f). Additionally, the γ-oscillation power in the RSC, AuD, and M2 significantly enhanced (Supplementary Fig. 2e, f).

LEC^{5} relays the RPE signal to the cortex

We conducted optrode recordings in the pan layer 5 driver line Rbp4-Cre mice (Fig. 4a–c and Supplementary Fig. 3a, b). Following the receipt of the reward in the US-only trials, the mean firing rates of eleven photo-identified Rbp4-positive units (well-isolated from six mice) increased significantly by 5.66 ± 0.59 Hz from the baseline. In comparison to the baseline of CS-only trials, there was no statistically significant increase in the firing rate during the CS period. However, as the training progressed, there was a notable increase in firing rate by 2.30 ± 0.55 Hz during the CS period, which was consistent with the firing pattern associated with RPE encoding in VTA^{DA}. We did not observe a distinct distribution of spike probability in Rbp4-positive cells over theta phases or low gamma phases through circular vector analysis (Supplementary Fig. 3c).

Given our previous research indicating a strong link between intercortical γ-synchrony and LEC^{5}, it is essential to investigate the mediating role of LEC^{5} in transferring the RPE signal from VTA^{DA} to the cortex[17]. In alignment with prior research, our anterograde trans-synaptic tracing experiments showed that cortical neurons downstream of LEC^{5} are predominantly located in the superficial layers, with only a limited subset of these downstream cells expressing parvalbumin (Supplementary Fig. 4).

To investigate afferent activity from LEC^{5} to the cortex, we recorded axonal calcium activities of LEC^{5}-RSC projections using a 2 P microscope (1466 boutons from seven mice, Fig. 4d, e), followed by noise reduction (Supplementary Fig. 5a, b). In the CS-only and US-only trials, 10.9% of the boutons were classified as CS-responsive, 20.8% as US-responsive, and 6.0% as CS/US co-responsive (Fig. 4f, g). Interestingly, a significant reduction in the mean activity of a considerable portion of the US-responsive boutons was observed during the final five US-only trials in comparison to the initial five trials, indicating habituation to the unexpected US presentation (Fig. 4h). Due to the limited photostability of untargeted GCaMP6s in axons, maintaining a stable field of view across multiple days is challenging. As a compromise, a swift experiment comprising 30 trials with varying US sizes was conducted (Fig. 4i), and the response window was substituted with direct reward delivery (exemption from triggering) to shorten imaging durations and prevent photobleaching. In the initial series of fifteen CS-US association trials, 25.5% of the boutons exhibited responses. In the following three series of trials—comprising five double US trials with two drops of water, five US omission trials without water delivery, and an additional set of five CS-US association trials with one drop of water—6.6% of boutons conformed to the TD learning model. The activity of these TD-boutons during the US period exhibited three distinct patterns in relation to the activity levels observed during the initial series of CS-US trials (0.086 ± 0.008, dF/F0): an elevation in activity when the reward was doubled (0.179 ± 0.013, dF/F0), a reduction when the reward was omitted (−0.014 ± 0.004, dF/F0), and a renewed increase upon the reintroduction of the reward (0.091 ± 0.012, dF/F0). Notably, this result cannot be attributed solely to the reflection of reward sizes rather than learning, as averaged responses in all four series of trials during the CS period were significantly larger than those in the preceding CS-only trials (Fig. 4j–m). Only 13.4% of TD-boutons exhibited responses during the CS-only phase. The majority of TD-boutons only developed responsiveness to the CS stimulus after associative learning. We classified these 97 TD-boutons into 70 independent axons based on distinct distribution patterns between intra-axonal and inter-axonal correlation coefficients (Supplementary Fig. 5c, d).

These data collectively suggest a bottom-up hierarchical cascade, whereby the RPE signal is initially conveyed from the VTA^{DA} to LEC^{5}, and culminating in an enhancement of intercortical γ-synchrony through common afferent excitatory projections from LEC^{5} (Fig. 4n).

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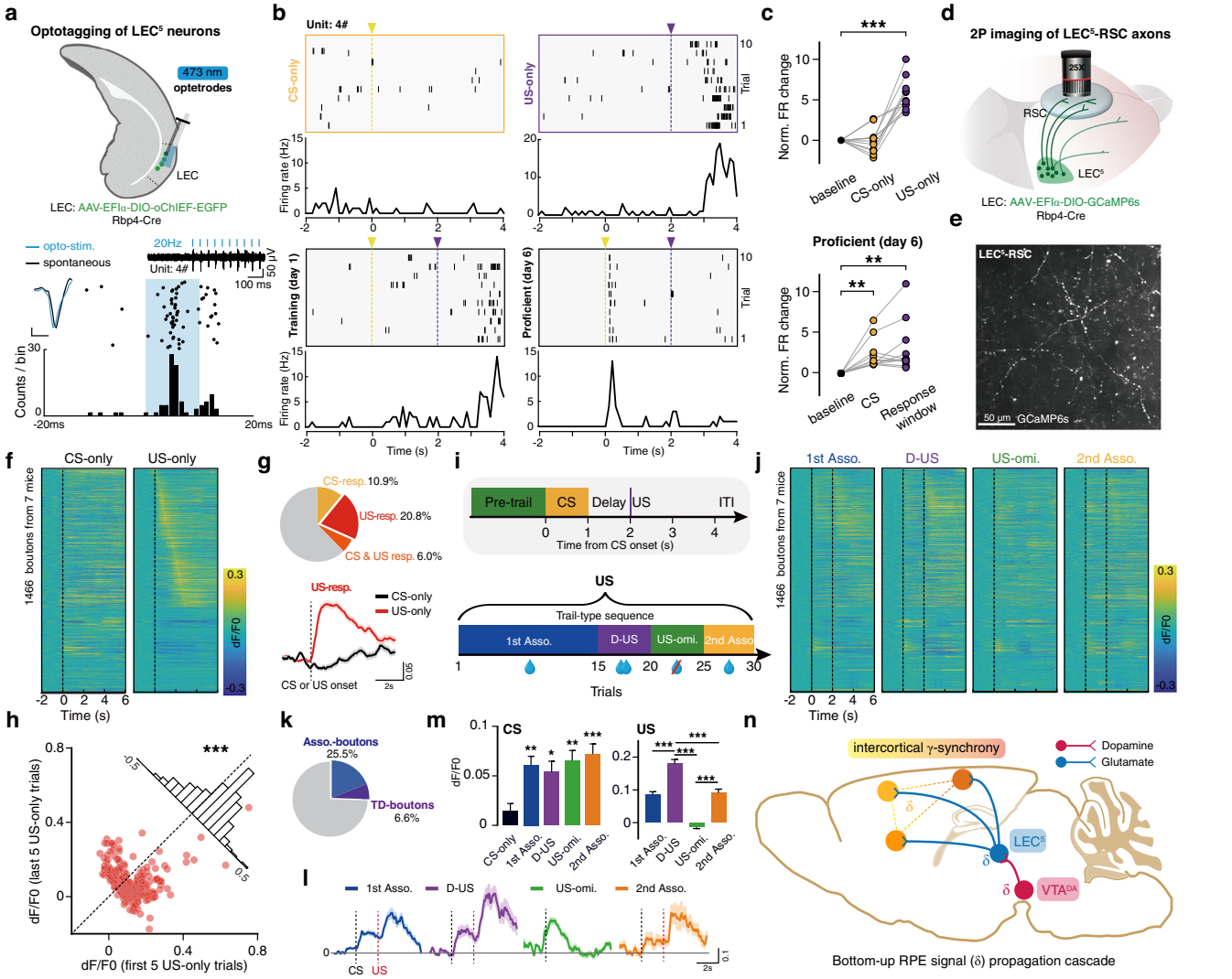


Fig. 4 | LEC⁵ relays the RPE signal to the cortex. a Top, viral and optotagging strategy for recording Rbp4-positive neurons. Bottom, the averaged laser-evoked (blue) and spontaneous (black) spike waveforms of a representative Rbp4-positive neuron, along with peristimulus time histograms of 20 Hz laser pulse stimulation. **b** Raster plots and firing rate changes of an example photoidentified LEC⁵ neuron across different stages of associative learning. The triangle and dashed line indicate the start for CS (yellow) and the response window (purple). **c** Top, changes in firing rates of photo-identified LEC⁵ neurons in CS-only and US-only trials. $P_{\text{baseline vs. US-only}} < 0.001$, $P_{\text{baseline vs. CS-only}} = 0.863$. Bottom, relative changes in their firing rates during the CS period and response window on day6. $P_{\text{baseline vs. CS}} = 0.002$, $P_{\text{baseline vs. Response window}} = 0.009$. $n = 11$ neurons. **d** Viral strategy for expressing GCaMP6s in LEC⁵ of Rpb4-Cre mice. A cranial window over RSC was installed. **e** Representative LEC⁵-RSC axon boutons from one of 7 mice. **f** Trial-averaged CS-only and US-only responses of LEC⁵-RSC boutons (1466 boutons from 7 mice), sorted by the peak response time to US. **g** Top, proportion of boutons that are CS-responsive, US-responsive, or co-responsive to both. Bottom, average calcium activities of the US-responsive group in response to either CS or US. Shading: S.E.M. **h** Scatter plot of the average activities of US-responsive boutons during the first five and the last five US-only trials. **i** Trial structures: 15 association trials (1 drop water; 1st Asso.), followed by 5 double-US trials (2 drops; D-US), 5 omission trials (no water; US-omi.), and finally 5 association trials (1 drop; 2nd Asso.). **j** Trial-averaged responses of LEC⁵-RSC boutons across different stages. **k** Proportion of the CS-US association responsive group and the TD-boutons. **l** Averaged responses of all 97 TD-boutons (form 7 mice) to four trial types. Shading: S.E.M. **m** Quantification of averaged responses within the CS period (left) or within 2 s after US delivery (right) of TD-boutons to all four trial types. $n = 97$ boutons. Error bar shows S.E.M. **n** Flatmap summary of the bottom-up RPE signal propagation cascade. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. *P* values were calculated using two-sided paired *t*-test (**c**), one-way ANOVA Tukey test (**m**), and two-sided Kolmogorov–Smirnov test (**h**). Source data are provided as a Source data file.

VTA^{DA}-LEC^{5} pathway facilitates associative learning

Numerous studies have established the causal role of VTA^{DA} as a TD learning signal in the brain. Activating or inhibiting VTA^{DA} can effectively influence the associative learning process in mice[19–24] and non-human primates[25]. To examine the necessity of the VTA^{DA}-LEC^{5} circuit in associative learning behaviors, we first conducted a loss-of-function experiment by time-locked optogenetic inhibition of eNpHR3.0-expressed VTA^{DA}-LEC dopamine projections. In each trial, a 589 nm laser was applied continuously for 3 s, covering the delay period and the response window during the entire associative training (Fig. 5a–c). Inhibition of VTA^{DA} projections to the LEC impaired the learning performance of eNpHR3.0-injected mice, in comparison to the mCherry control group (Fig. 5f). A comparison of the relative PLV change during the CS period on day6 between the inhibition group and the control group showed that the levels of intercortical γ-synchrony significantly increased by 34.78 ± 4.93% in the control group, while it remained unchanged (−1.13 ± 2.04%) in the inhibition group (Fig. 5d, e and Supplementary Fig. 6a).

Subsequently, we activated LEC^{5} while simultaneously suppressing upstream VTA^{DA}-LEC projections in DAT-Cre × Rbp4-Cre mice during the entire associative training (Fig. 5g–i). Mice in the

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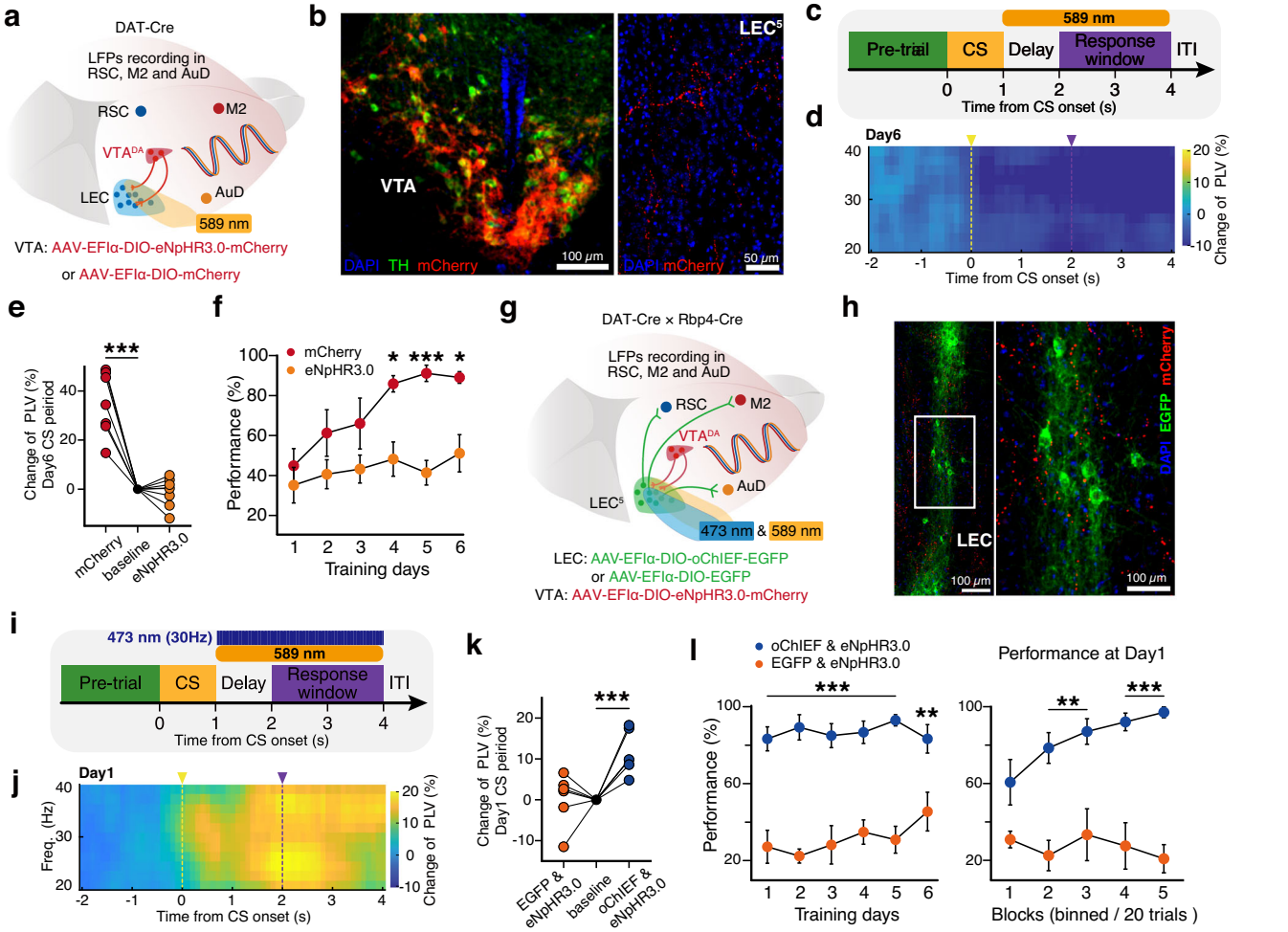


Fig. 5 | Facilitation of learning and memory retention through the manipulation of the VTA^{DA}-LEC^5 circuit. a Viral strategy for the bilateral expression of eNpHR3.0-mCherry or mCherry in VTA^{DA} of DAT-Cre mice. Optical fibers were bilaterally implanted above the LEC. LFPs in RSC, M2, and AuD were recorded simultaneously. **b** Left, representative Cre-mediated expression of eNpHR3.0-mCherry in VTA^{DA} from one of 8 mice. Dopamine neurons were stained with Tyrosine Hydroxylase (TH) antibody. Right, representative VTA^{DA} axons in LEC. **c** Trial structure of CS-US training. In each trial, a 589 nm laser was applied continuously for 3 s, covering both the delay period and the response window. **d** Representative trial-averaged intercortical γ-synchrony spectrogram of the inhibition group on day6. **e** Quantification of the changes in averaged PLV within the CS period in the mCherry control group ($P = 0.0004$, $n = 7$ mice) and the inhibition group ($P = 0.596$, $n = 8$ mice). Compared to each of their respective averaged baseline (within 2 s before CS) on day6. **f** Performance of the mCherry control group ($n = 7$ mice) and the inhibition group ($n = 8$ mice). $P_{day4} = 0.017$, $P_{day5} < 0.001$, $P_{day6} = 0.014$. Error bar shows S.E.M. **g** Viral strategy for the bilateral expression of eNpHR3.0-mCherry in VTA^{DA} and oChIEF-EGFP or EGFP in LEC^5 in DAT-Cre × Rpb4-Cre mice. Optical fibers were bilaterally implanted above the LEC. LFPs in RSC, M2, and AuD were recorded simultaneously. **h** Representative Cre-mediated expression of oChIEF-EGFP in LEC^5 and eNpHR3.0-mCherry in VTA^{DA} axons from one of 7 mice. **i** Trial structure of CS-US training. In each trial, a 589 nm (constant) and a 473 nm (30 Hz) hybrid laser were applied simultaneously for 3 s, covering the delay period and response window. **j** Representative trial-averaged intercortical γ-synchrony spectrogram from the oChIEF & eNpHR3.0 group on day1. **k** Quantification of the changes in averaged PLV within the CS period in the EGFP & eNpHR3.0 control group ($P = 0.919$, $n = 6$ mice) and the oChIEF & eNpHR3.0 group ($P < 0.001$, $n = 7$ mice), compared to their respective averaged baseline (within 2 s before CS onset) on day1. **l** Performance of the EGFP & eNpHR3.0 control group ($n = 6$ mice) and the oChIEF & eNpHR3.0 group ($n = 7$ mice) over six training days (left) and five blocks on day1 (right). Error bar shows S.E.M. $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. *P* values were calculated using two-tailed paired *t*-test (**e**, **k**), and a two-way ANOVA with Sidak's post hoc multiple comparisons test (**f**, **l**). Source data are provided as a Source data file.

experimental group showed significantly better performance compared to the group that received solely VTA^{DA}-LEC inhibition (Fig. 5l, left). Strikingly, the performance disparities between the groups were discernible even in the initial stages of the training process, indicating accelerated learning (Fig. 5l, right). Furthermore, the relative change in PLV during the CS period on day1 between the groups showed that intercortical γ-synchrony increased significantly (11.24 ± 1.84%) in the experimental group mice, while it remained unchanged (−0.28 ± 2.60%) in the inhibition-only group (Fig. 5j, k and Supplementary Fig. 6b). Notably, the increase in PLV observed during the delay and response intervals was attributable to laser stimulation, whereas the enhancement within the CS period was endogenous (Fig. 5j).

Stimulation of LEC^5 improves contextual memory retention

To examine whether LEC^5-mediated intercortical γ-synchrony improves learning and memory retention in more complex tasks beyond CS-US associative trials, we conducted both positive and negative conditioning assessments involving contextual associations: conditioned place preference (CPP) and contextual fear conditioning (CFC), both of which require continuous sensory feedback during environmental exploration (Fig. 6a, d).

The initial day of the CPP experiment demonstrated that mice subject to dietary restrictions exhibited a preference for the dark side of the light-dark box. On the second day, we separated the two boxes and allowed the mice to explore for 10 min. Mice were exposed to 30 Hz laser stimulation throughout foraging in the isolated light box

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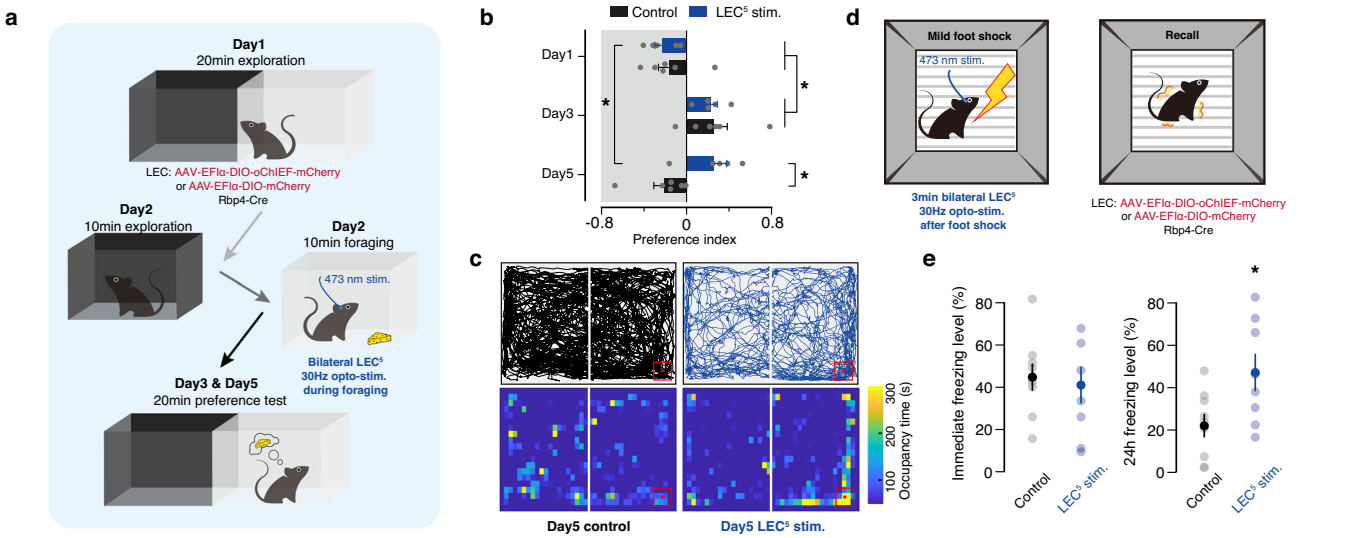


Fig. 6 | Stimulation of LEC^5 improves contextual memory retention. **a** Procedures of the CPP experiment. AAV-DIO-oChIEF-mCherry or AAV-DIO-mCherry was injected into bilateral LEC^5 in Rpb4-Cre mice. Optical fibers were bilaterally implanted above the LEC. Laser stimulation at 30 Hz was administered during the foraging session in the dark box on day2. **b** Quantification of the preference index for the oChIEF group ($n = 5$ mice) and the control group ($n = 6$ mice). $P_{\text{Control-day3 vs. Control-day1}} = 0.037$, $P_{\text{LECstim-day3 vs. LECstim-day1}} = 0.043$, $P_{\text{LECstim-day5 vs. LECstim-day1}} = 0.025$, $P_{\text{LECstim-day5 vs. Control-day5}} = 0.026$. Each dot represents one mouse. Error bar shows S.E.M. **c** Representative trajectory (top) and occupancy map (bottom) during memory recall on day5 for the control group and LEC^5 stimulation group (right), respectively. **d** Paradigm of CFC training and recall tests. Mice received a mild electric foot shock (0.3 mA, 2 s) during training. The freezing levels were measured twice: once immediately after the training and again 24 h later. **e** Quantification of the freezing levels in the mCherry control group ($n = 9$ mice) and the oChIEF group ($n = 8$ mice) for immediate recall (left, $P = 0.670$) and 24-h recall (right, $P = 0.024$), respectively. Each dot represents one mouse. The dark dot indicates the average value. Error bar shows S.E.M. *$P < 0.05$. P values were calculated using two-way ANOVA Tukey test (**b**), and two-tailed t-test (**e**). Source data are provided as a Source data file.

with buried food. In the subsequent test conducted on day3, the light and dark compartments were interconnected, and both the oChIEF group and the mCherry group of mice spent more time exploring the bright side of the box, where food had previously been buried. However, in the test conducted on day5, only the oChIEF group mice continued to spend more time exploring the bright side, indicating stable long-term memory retention. In contrast, the mCherry group of mice reinstated their preference for dark environments (Fig. 6b, c).

In the CFC experiment, mice were subjected to a mild electrical foot shock (0.3 mA, 2 s), followed by 3 min of optogenetic stimulation of LEC^5. There were no significant differences between the mCherry control group ($n = 9$) and the oChIEF group ($n = 8$) in travel distance during the 3 min habituation period or in running speed during the foot shock (Supplementary Fig. 6c, d). No observable difference in freezing levels was noted between the oChIEF group (40.56 ± 8.42%) and the mCherry group (45.02 ± 6.13%) during the immediate recall of fear memory. However, 24 hours later, only the oChIEF group preserved elevated freezing levels (46.98 ± 8.59%) during the second recall test, indicating a robust fear memory retention (Fig. 6e). Given its bidirectional enhancement of both positive and negative memory, the VTA^{DA}-LEC^5 circuit may contribute to the computation of contextual prediction errors within the neocortex.

Intercortical γ-synchrony entrains the activities of the latent RSC Engrams

To determine whether intercortical γ-synchrony is a cause or a consequence of cortical neural activities, it is essential to understand its function from a cellular perspective. We recorded the calcium activities of RSC neurons using a two-photon (2 P) microscope, while concurrently employing custom-made 2-shank ultraflexible electrode arrays[26] or tungsten electrodes, to record LFPs from both the RSC and AuD regions (Fig. 7a–c and Supplementary Fig. 7a–d). Calcium activities of a total of 685 neurons from eight mice were recorded during the CS-only and US-only stimulation epochs on day1, during the CS-US association learning epoch on day3, and during the memory recall epoch on day7 (Fig. 7d and Supplementary Fig. 7e, f). The trial-averaged responses observed in the initial experiments indicated that a certain percentage of cells responded to the CS stimulus (4.67%), the US stimulus (34.01%), and both the CS and US stimuli (9.49%). Excluding two CS-only encoding neurons and 17 US-only encoding neurons throughout the entire experiment, 93 neurons exhibited responses to either the CS, the US, or both during the association training on day3. These neurons were categorized within the association-encoding group (Asso.). On day7, all mice demonstrated proficient performance, with 78 neurons exhibiting activity confined to narrow windows within the CS period. We classified these neurons as recall group cells (Recall). Next, we retraced the activity of these Recall cells on day3 and found that 18 cells (2.63% of the total) had previously been categorized within the Asso. group, exhibiting transient activity during both the learning and recall stages (Fig. 7e). We classify these 18 cells as latent cortical Engrams that maintain associative memory.

In mice implanted with ultraflexible electrode arrays, it was observed that the selection of either deep or superficial oscillations influenced the calculation of γ-synchrony (Supplementary Fig. 7c). Consistent with experiments employing tungsten electrodes inserted into the superficial cortical layer, superficial LFPs were selected for the computation of RSC-AuD γ-synchrony. In comparing the correlations between RSC single-cell calcium activities and the concurrent RSC-AuD pairwise PLV change on day3, the correlation coefficient for the RSC latent Engram group was significantly higher than that for the non-Engram group (Fig. 7f–h). Crucially, the increase in PLV preceded the rising time point of calcium activities (0.21 ± 0.13 s) in the RSC latent Engrams on day3, indicating that γ-synchrony entrains the activities of the latent RSC Engrams during learning. Given the unique projections of LEC^5 to various cortical regions, we hypothesize that mechanisms similar to those observed in the RSC may also exist in other areas. Nonetheless, a notable decrease in the time lag (−0.15 ± 0.05 s)

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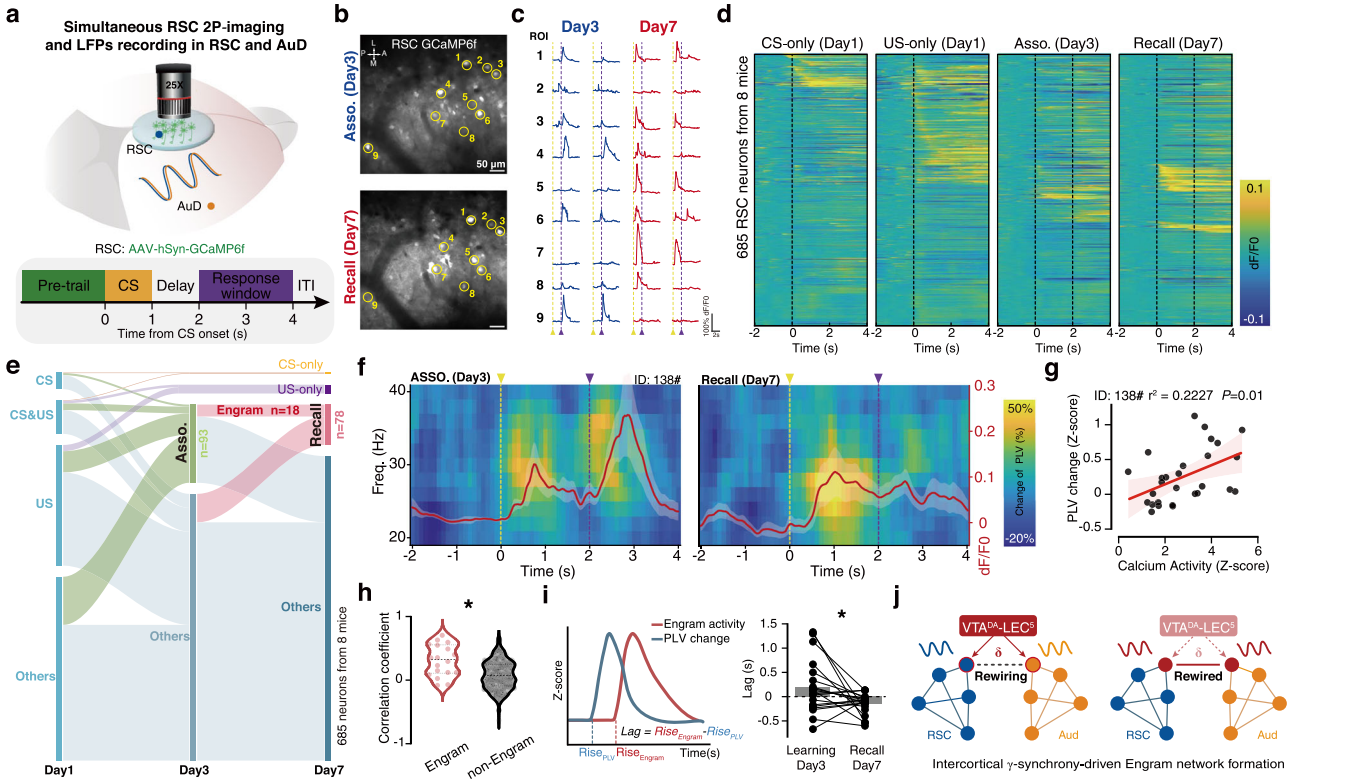


Fig. 7 | Intercortical γ-synchrony entrains the activity of RSC latent Engrams. **a** Top, viral strategy for expressing GCaMP6f in RSC. LFPs in RSC and AuD were recorded simultaneously. Bottom, trial structure for CS-US training and recall from day2 to day7. **b** Representative 2 P field of view during training (top) and recall (bottom) from one of 8 mice. **c** Calcium transient traces of day3 (blue) and day7 (red) for the numbered ROI in (**b**). The triangle and dashed line indicate the onset times for CS (yellow) or US (purple) delivery. **d** Trial-averaged activities of all 685 neurons ($n = 8$ mice) across different stages, sorted by response to CS-only, US-only, association training (day3), and memory recall (day7) in succession. **e** An alluvial diagram illustrating the grouping trends of response characteristics across different stages for all neurons. **f** Averaged RSC-AuD γ-synchrony spectrogram with activity of one representative RSC latent Engram during learning (left) and recall (right). **g** Trial-by-trial correlation between calcium activity of a representative latent Engram (same neuron in **f**) and the changes in intercortical γ-synchrony on day3. Shading: 95% confidence bands. **h** Quantification of the trial-by-trial correlation coefficient between the changes in intercortical γ-synchrony and calcium activities in each latent Engram or the remaining non-Engram neurons. (**i**) Left, method for calculating the rising time lags between intercortical γ-synchrony (blue) and calcium activities (red) of the latent Engram. Right, quantification of time lags during learning (day3) and recall (day7). $P = 0.023$. $n = 18$ neurons. **j** Diagram summarizing the underlying mechanism of intercortical γ-synchrony involved in RPE signal propagation and cortical Engram network formation. $^*P < 0.05$. P values were calculated using the two-sided Pearson correlation test (**g**), two-sided Kolmogorov–Smirnov test (**h**), and two-tailed paired t-test (**i**). Source data are provided as a Source data file.

between calcium activity of these latent RSC Engrams and the increase in PLV was observed on day7 (Fig. 7i). In addition to the correlation study, we conducted a loss-of-function experiment by selectively expressing hM4D in bilateral LEC[5] of Rbp4-Cre mice. Chemicalgenetics inhibition of LEC[5] on day8 did not influence the activity of Engram populations nor impair the behavioral performance of the mice (Supplementary Fig. 9).

Based on these results, we suggest that the LEC-mediated γ-synchrony signal precedes the activation of the latent Engram during the initial phases of learning. Conversely, once the mice reach the proficient level, the activation of the Engram network occurs before the emergence of the intercortical γ-synchrony signal, indicating an intercortical γ-synchrony-mediated mechanism for the maturation of the cortical memory networks (Fig. 7j).

Enhancement of γ-synchrony in the human brain during prediction error processing

The dopamine-driven RPE teaching signal has been observed in several species, including Drosophila[27], zebra finches[28], and humans[29], demonstrating its conserved feature across species. Furthermore, neural representations of broadly defined prediction errors (discrepancies between expectations and perceptions) have recently extended beyond cue-reward associations to include areas such as social interaction[30,31], threat avoidance[32], addiction[33], pain expectation[34], and speech perception[35], highlighting the functional diversity of prediction error signals in the brain.

We performed a reading task with patients undergoing awake glioma resection with language mapping (Fig. 8a, b). High-density ECoG grids were placed intraoperatively over the temporal, frontal, and parietal lobes to record LFPs, and a subset of 30 channels per patient was selected for PLV calculation (Supplementary Fig. 8a and Supplementary Tables 3, 4). Each trial consisted of three sequential phases: an initial fixation period of 1 s, during which a central crosshair was displayed; a perception phase of 2 s, during which two-character Mandarin phrase stimuli were presented on the screen; and a subsequent reading phase of 1.5 s. Sound waves during articulation were recorded to verify an accurate reading. The stimuli comprised three real-phrases and three pseudo-phrases. Each phrase was reiterated three times in random sequence within a single block, with several minutes of rest between the two blocks, culminating in a total of six repetitions per phrase. For the statistical analysis, we specifically selected electrode pairs that exhibited at least a 5% increase in PLV during pseudo-phrase presentation in the first block (40.68% frontal-frontal, 41.37% frontal-parietal, 42.90% frontal-temporal, 39.90% parietal-parietal, 35.98% parietal-temporal, and 39.88% temporal-temporal pairs exceeded the 5% threshold). Data from these pairs

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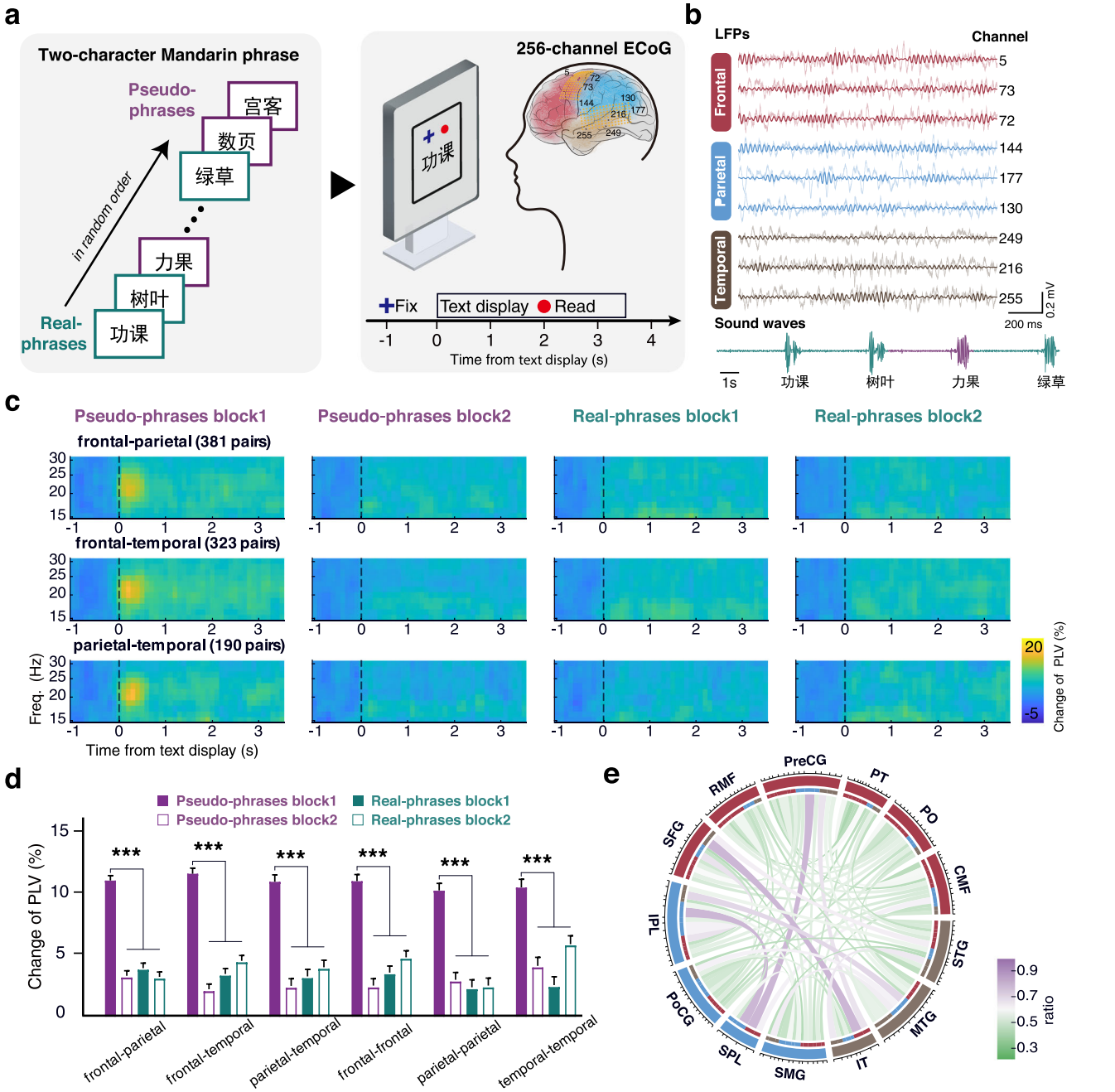


Fig. 8 | Unexpected pseudo-phrases sensitive γ-synchrony in the human brain. **a** Left, two-character Mandarin pseudo-phrases (purple) and real-phrases (green) were presented to the participant in a random order. The experiment consisted of two blocks, with each phrase appearing three times per block. Right, each trial consisted of a 1 s warning cue, a 2-s text presentation, and a subsequent reading prompt. Each participant used two 128-channel ECoG electrode arrays covering major areas, including the frontal lobe (red), temporal lobe (brown), and parietal lobe (blue). The human brain illustration is adapted from https://www.magnific.com/free-vector/human-brain-illustration-isolated-white-background-side-view-cartoon-brain-vector-image_89163136.htm. **b** Representative gamma band (20–40 Hz, dark lines) waves extracted from ECoG raw traces (tint lines) in the frontal lobe, parietal lobe, and temporal lobe, respectively. Sound waves were recorded to verify an accurate reading. **c** Spectrograms of γ-synchrony of averaged frontal-parietal, frontal-temporal, and parietal-temporal pairs exceeding the threshold for pseudo-phrases and real-phrases during the 1st and 2nd blocks, respectively. (**d**) Quantification of changes in γ-synchrony within 0.5 s after the text display across brain regions in each group, for all participants, all phrases, and all pairs. ***$P < 0.0001$. P values were calculated using one-way ANOVA Dunnett test. $n_{frontal\text{-}parietal} = 1524$, $n_{frontal\text{-}temporal} = 1292$, $n_{parietal\text{-}temporal} = 760$, $n_{frontal\text{-}frontal} = 856$, $n_{parietal\text{-}parietal} = 672$, $n_{temporal\text{-}temporal} = 528$ sets of data. Error bar shows S.E.M. **e** Chord diagram summarizing the connection of elevated γ-synchrony between and within three brain lobes based on the ratio of pairs that exceed the PLV increase threshold during the first block of pseudo-phrases, out of the total number of pairs calculated. Attributes of color hue are used to distinguish different brain lobes (red: frontal lobe; blue: parietal lobe; brown: temporal lobe). The colorbar and the width of connecting lines are proportionally sized according to the ratio of pairs that exceed the threshold. Caudal middle frontal, CMF Inferior parietal lobule, IPL Inferior temporal gyrus, IT Pars opercularis, PO Postcentral gyrus, PoCG Precentral gyrus PreCG Pars triangularis, PT Rostral middle frontal gyrus, RMF Superior frontal gyrus, SFG Supramarginal gyrus, SMG Superior parietal lobule, SPL Superior temporal gyrus, STG Source data are provided as a Source data file.

showed a broader phenomenon of elevated intercortical synchrony (20–25 Hz) between and within three brain lobes in the 1st block of pseudo-phrases, indicating cortical processing of prediction errors. Furthermore, the enhancement in synchrony diminished in the subsequent block, suggesting rapid familiarization with pseudo-phrases based on knowledge acquired in the first block (Fig. 8c, d and Supplementary Fig. 8b). Further analysis across brain subregions showed a greater likelihood of increased synchrony in connections such as precentral gyrus to superior parietal lobule (PreCG-SPL), inferior parietal lobule to superior parietal lobule (IPL-SPL), and superior frontal gyrus to inferior temporal gyrus (SFG-IT) (Fig. 8e). Additionally, no significant alterations in the power of γ-oscillations within the parietal lobe region were observed (Supplementary Fig. 8c).

Discussion

In this study, we observed a remarkable resemblance between the patterns of intercortical γ-synchrony and the dopamine-driven RPE signal in both mice and humans, indicating that the neural signature of the RPE signal within cortical areas is represented as intercortical γ-synchrony. Growing evidence shows that RPE-encoding cell populations exist in the neocortex[36–38]. However, it remains unclear how these higher regions update current predictions when they no longer match the real-world conditions. Our study reveals a previously unknown circuit architecture that transmits the midbrain dopamine RPE signal to the extensive cortical regions.

LEC has been documented to participate in temporal segmentation of episodic events[39], and reward learning[40,41]. Two studies from the Igarashi laboratory clarify the pivotal function of VTA projections to the fan cells within the LEC layer2a, as well as the bidirectional loop circuit between the medial prefrontal cortex and the deep-layer LEC, in learning odor-based associations[42,43]. Notably, the entorhinal region has also been identified as a primary area of vulnerability in the early stages of Alzheimer's disease[44]. In line with these advances in LEC research, our study deepens the understanding of LEC's role in the processing of the RPE signal. The RPE-driven retrospective reinforcement mechanism enables animals to infer the specific actions that elicit dopamine release[45]. Such associations constitute the fundamental framework of memory formation. We confirmed that the bottom-up VTA^{DA}-LEC^5 RPE signal cascade is not only essential but also sufficient for supporting associative learning and memory retention. Fine-tuning LEC^5 activity may facilitate the differentiated development of specialized neural networks tailored to specific memories.

The relationship between RPE and memory encoding has been conceptualized through two distinct mechanisms. The signed reward prediction error (SRPE) model asserts that memory is selectively enhanced by positive deviations from expectation, implicating outcome valence as a critical modulatory factor. By contrast, the unsigned reward prediction error (URPE) model decouples memory strength from valence, arguing instead that the magnitude of surprise—quantified as the absolute difference between predicted and experienced outcomes—serves as the primary determinant of memory encoding efficacy[46]. Our experimental results align with the latter framework: optogenetic activation of LEC^5 concurrently augments spatial memory for reward locations and contextual fear memory. This cross-domain facilitation indicates that LEC^5-mediated memory enhancement operates in a manner of unsigned prediction error.

Given the ambiguity regarding the origin and significance of LFPs, despite extensive research into the relationship between neural oscillations and cognitive functions, the spectral pattern of a particular oscillation is often questioned as an epiphenomenon that merely reflects the dynamic mode of neural networks rather than acting as a causal factor in specific neural activities[47]. At the single-cell level, we discovered that instantaneous intercortical γ-synchrony functions as a reference point for entraining the activities of RSC latent Engrams during reinforcement learning at the early stage. Neuronal spike probability in a particular phase of oscillation can be modulated by the extracellular electric field, which may orchestrate the spike timing of distributed cell ensembles and guide the formation of intercortical connections[48,49]. Considering that γ-oscillations are more localized in time and space[47], our data indicate that common inputs from the LEC^5 facilitate the integration of functionally segregated cortical regions through intercortical synchrony at the low gamma band. Notably, we observed that cortical Engram activity precedes the increase in intercortical γ-synchrony during memory recall, and the activity of the Engram population was independent of LEC^5, indicating a mechanism mediated by intercortical γ-synchrony for the development of cortical memory networks in the learning phase.

Entrainment to a specific frequency of oscillation is cell-type dependent. Inhibitory neurons play a crucial role in regulating γ-oscillations[50,51]. Further investigation is required to elucidate the mechanisms underlying the local processing of afferent information from LEC^5 within the cortical microcircuit and to understand its implications in intercortical connections.

Methods

Mice

All mice were housed in the Animal Core Facility at ShanghaiTech University and kept on a 12 h light/dark cycle with free access to food and water. The facility maintained a steady temperature of 22–26 °C, relative humidity between 40% and 70%, and noise levels below 60 dB. Both male and female mice aged 4–6 months were used (see Supplementary Table 5). Mice of different sexes are not treated differently across experimental methods. C57BL/6 J mice were purchased from Shanghai Jihui Laboratory Animal Care. The Rbp4-Cre (MMRRC, Stock No. 031125-UCD) line was obtained from the Mutant Mouse Resource & Research Centers (MMRRC). The Ai9 (JAX, Stock No. 007909) line was obtained from the Jackson Laboratory. The DAT-Cre (JAX, Stock No. 006660) line was kindly provided by Dr. Ji Hu (ShanghaiTech University). To generate DAT-Cre × Rbp4-Cre double-transgenic mice, the DAT-Cre line was bred with the Rbp4-Cre line. All animal experiments were conducted in accordance with protocols approved by the Institutional Animal Care and Use Committee of ShanghaiTech University (20241226001).

Stereotaxic surgery

Mice were anesthetized with 5% isoflurane for induction and placed in a stereotaxic apparatus (RWD Life Science Co., Ltd., China), where they were maintained under 1% isoflurane anesthesia. The scalp was disinfected with povidone-iodine, shaved, and incised to expose the skull. The soft tissue above was carefully removed, and the skull was precisely leveled by ensuring the Bregma and Lambda points were aligned on the same dorsal-ventral plane for further procedures. All coordinates are provided relative to Bregma. For simplicity, all injection or implantation coordinates used in the following experiments are listed here: RSC (AP: −3.0 mm, ML: −0.5 mm, DV: −0.3 mm); M2 (AP: −1.0 mm, ML: −1.0 mm, DV: −0.3 mm); AuD (AP: −3.0 mm, ML: −4.0 mm, DV: −0.3 mm); VTA (AP: −3.08 mm, ML: −0.6 mm, DV: −4.5 mm); LEC (AP: −4.0 mm, ML: −4.2 mm, DV: −4.2 mm). All the viruses (titers varied from 1 to 2×10^{13} ml^{-1}) were purchased from Obio Technology (Shanghai) Co., Ltd. A glass pipette (Wiretreol II 5 and 10, Drummond) was beveled at a 45-degree angle and filled with mineral oil; it was subsequently connected to an automated hydraulic pump (RWD Life Science Co., Ltd., China) for virus injection. Optical fibers (200 μm core diameter, NA 0.37, HealthiGlobal Tech Co., Ltd., China) were employed for fiber photometry or optogenetics manipulation experiments. In experiments involving head-fixed mice, all implants and a custom-designed titanium head-plate were affixed to the skull using dental cement. Postoperative recovery for the mice was conducted on a heated platform until they exhibited normal behavior. Within one week, they were provided with fortified nutritional jelly containing the

analgesic carprofen (5 mg). A minimum duration of two weeks was designated for recovery before advancing to any subsequent phase of the experiment.

For multi-site electrode implantation, custom-made electrodes with PFA-coated tungsten wires (A-M Systems, 795500) were used to record LFPs. Only electrodes with an impedance below 2 MΩ were employed. Craniotomies were performed at RSC, AuD, and M2, followed by stereotaxic implantation. Two skull screws were implanted over the cerebellum, one acting as a reference electrode and the other as a ground electrode.

For the fiber photometry experiment, AAV9-EF1α-DIO-GCaMP6f (150 nL) was injected into the unilateral VTA of DAT-Cre mice at a rate of 50 nL/min. After the injection, an optical fiber was implanted above the injection site. The procedure for tungsten electrode implantation and the recording coordinates in RSC, AuD, and M2 were consistent with those described above.

For the optogenetic stimulation of the VTA^{DA} experiment, a separate group of DAT-Cre mice was injected with AAV9-EF1α-DIO-oChIEF-EGFP (150 nL) into the unilateral VTA. In addition to RSC, AUD, and M2, a tungsten electrode was also implanted in the ipsilateral LEC. The procedures for implanting the tungsten electrode and optical fiber were consistent with those described above.

For the optogenetic tagging experiment, AAV9-EF1α-DIO-oChIEF-EGFP (150 nL) was injected into the unilateral LEC^5 of Rbp4-Cre mice. Subsequently, four skull screws were attached, with one serving as a ground and another as a reference. A 32-channel optrode array, assembled from one optical fiber and eight tetrodes arranged in a 2 × 4 configuration, spaced 150 µm apart (KedouBC, China), was then angled laterally at 10 degrees and inserted at the LEC^5 injection site. The exposed cortical surface was carefully covered with surgical silicone adhesive (Kwik-Sil, World Precision Instruments, USA).

For the anterograde trans-synaptic tracing experiment, AAV1-hSyn-Cre (150 nL per hemisphere) was injected into the bilateral LEC of Ai9 mice.

For the optogenetic inhibition of the VTA^{DA}-LEC^5 experiment, AAV9-EF1α-DIO-eNpHR3.0-mCherry (150 nL per hemisphere) or AAV9-EF1α-DIO-mCherry (150 nL per hemisphere) was injected into the bilateral VTA of DAT-Cre mice. Subsequently, two optical fibers were bilaterally implanted above the LEC. Tungsten electrodes were also implanted in the RSC, AuD, and M2 areas of the right hemisphere.

For the combined optogenetic inhibition of the VTA^{DA}-LEC^5 pathway and stimulation of the LEC^5 experiment, AAV9-EF1α-DIO-eNpHR3.0-mCherry (150 nL per hemisphere) was bilaterally injected into the VTA of DAT-Cre × Rbp4-Cre mice. Additionally, bilateral LEC^5 were injected with either AAV9-EF1α-DIO-oChIEF-EGFP (150 nL per hemisphere) or AAV9-EF1α-DIO-EGFP (150 nL per hemisphere). Subsequently, two optical fibers were implanted bilaterally above the LEC. Tungsten electrodes were also implanted in the RSC, AuD, and M2 areas of the right hemisphere.

For the CPP and CFC experiments, Rbp4-Cre mice received bilateral injections into LEC^5 of either AAV9-EF1a-DIO-oChIEF-mCherry (150 nL per hemisphere) or AAV9-EF1a-DIO-mCherry (150 nL per hemisphere). After the injections, two optical fibers were implanted bilaterally above the LEC^5.

For 2 P imaging of LEC^5-RSC axons, AAV9-EF1α-DIO-GCaMP6s (150 nL) was injected into the unilateral LEC^5 of Rbp4-Cre mice. A 4–5 mm circular craniotomy was performed centered on the RSC using a high-speed drill and a dissecting microscope for visualization. Subsequently, the craniotomy was sealed with a 5-mm-diameter glass coverslip (World Precision Instruments, USA). 2 P calcium imaging was performed one month after the surgery.

For simultaneous 2 P imaging of RSC neurons and LFP recording in the RSC and AuD, a 4–5 mm circular craniotomy was performed over both regions. An injection of AAV9-EF1α-GCaMP6f (150 nL) was delivered into the RSC at a depth of 300 µm below the pial surface. Next, the 2-shank ultraflexible electrode arrays were implanted in both the RSC and AuD ($n = 4$). The design and configuration of the 2-shank ultraflexible electrode arrays are shown in Supplementary Fig. 7a, b. The implantation procedure adhered to protocols described in the literature[52]. For mice using tungsten electrodes to record LFPs from the superficial layers of the RSC and AuD ($n = 4$), ultra-fine tungsten wires (PTFE-coated, 20 µm in diameter, California Fine Wire, 100211) were manually bent into electrodes approximately 0.3 mm long and inserted into the cortex as shown in Supplementary Fig. 7d. For chemogenetic inhibition experiments in Rbp4-Cre mice, AAV9-EF1α-GCaMP6f (150 nL) was delivered into the RSC at a depth of 300 µm below the pial surface, and AAV9-EF1α-DIO-hM4Di-mCherry (150 nL per hemisphere) was bilaterally injected into the LEC. The craniotomy was then sealed with a 5-mm-diameter glass coverslip. 2 P calcium imaging was conducted at least one month after surgery.

Human participants

This study involved eight native Mandarin speakers (five males and three females; aged 25–60 years) with left-hemisphere language dominance who underwent awake brain tumor resection with language mapping at Huashan Hospital, Shanghai, China. All participants were clinically indicated for awake surgery. High-density electrocorticography (ECoG) grids were placed intraoperatively over the temporal, frontal, and parietal cortices to record local field potentials (Supplementary Fig. 8 and Supplementary Table 3). The grid was positioned by an experienced neurosurgeon, guided by clinical expertise and the necessity to avoid the tumor. The study was approved by the Institutional Review Board (IRB) of Fudan University Huashan Hospital (HIRB KY2019-538) and conducted in strict compliance with ethical standards. All participants were informed of the voluntary nature of the study and their right to withdraw at any point, and written consent was obtained from each.

The pial surfaces were reconstructed from preoperative T1-weighted MRI scans using FreeSurfer (v7.2.0). The intraoperative localization of electrode grids was assisted by the Medtronic StealthStation S7 neuronavigation system, which recorded the three-dimensional coordinates of the grid corners. These fiducial markers were aligned to the preoperative MRI space, supplemented by intraoperative photographs for spatial validation. The remaining electrode centroids were calculated with the Python "img_pipe" package (v1.3), which uses geodesic interpolation algorithms based on the established corner coordinates. All electrode positions were normalized to the MNI 152 asymmetric template (2009 release) through nonlinear volumetric registration. Anatomical labeling assigned each electrode to a specific cortical region according to the Desikan-Killiany atlas.

CS-US associative learning

Mice were subjected to a water-restricted diet to maintain their body weight at 80–85% of their initial weight. Training started once their weight reached 85%. They were monitored daily for overall health during the experiment. Mice were head-fixed in a custom frame for training. The day before training, a shaping session was conducted, during which mice received 10 CS-only (2 kHz tone) trials and 10 US-only (one drop of water, ~7 µL) trials, delivered separately with a randomized 5–8 s inter-trial interval (ITI). Afterwards, mice performed the CS-US associative learning task (100 trials per day). Each trial included a 4–6 s lick withhold period, a 1 s CS delivery, a 1 s delay, and a 2 s response window. Only licking the water pipe within the response window activates the infrared sensor to dispense a drop of water as a reward. The ITI duration is randomly set between 5 and 8 s.

The CS replacing experiment was conducted on day7 with six proficient mice from the earlier CS-US association training. In this experiment, a blue light cue was presented 1 s before the tone cue (Fig. 2a).

For the US-size fluctuation experiment, the CS-US contingency was adjusted by randomly changing the US size in each trial on day7 for the other seven proficient mice (Fig. 2e). The trial structure followed the standard CS-US task, except that the US size during the response window varied across three trial types: US-omission (no reward), regular US size with one drop of water (~7 μL), and double US size with two drops of water (~14 μL). A total of 120 trials were conducted in a 1:1:1 randomized ratio. The trial-by-trial relative changes in RPE values after delivering different rewards to mice were calculated using the TD learning iterative algorithms. Trials where animals did not water-lick at all from CS to ITI were excluded, as these trials indicated incomplete engagement with the environment. The initial expectation of reward (VI) for each mouse was set to its hit rate on day6, with a learning rate (α) of 0.5. The outcome (r_t) for different US sizes in each trial was assigned a value of 1.5 for two drops of water, 1 for one drop of water, and 0 for reward omission.

The CS-US associative learning task was performed in a custom-made behavioral box controlled by an Arduino microcontroller. The interior surfaces of the box are covered with sound-absorbing foam to reduce environmental noise. In addition to an infrared video camera that monitored animal behavior, an infrared photobeam sensor provided millisecond-precise detection and timing of licking events. Performance for each mouse was assessed as the fraction of rewarded trials out of all training trials completed that day. The behavioral data were hardware-synchronized with electrophysiology recording, fiber photometry, and 2 P imaging.

LFPs recording and analysis

The multi-channel data were collected and amplified using an Apollo neural data acquisition system (v1.0, Bio-Signal Technologies, USA). The inner walls of the behavioral box were lined with a continuous copper mesh connected to ground to shield against electromagnetic interference. LFPs were recorded at 1 kHz and digitally stored after online band-pass filtering between 1 and 200 Hz. Unless stated otherwise, all analyses were conducted with custom MATLAB scripts. Before further analysis, channels with low signal-to-noise ratios (SNR) were identified and excluded. Causes of low SNR included 50 Hz power-line interference, electromagnetic noise from nearby equipment, and poor contact between the recording screw and cortical tissue. Power-line artifacts were reduced using the rmlinesc function from the Chronux toolbox (https://chronux.org). Zero-phase band-pass filtering was performed using MATLAB-designed IIR filters combined with the built-in filtfilt function. LFP analyses in this study focused on three metrics: LFP power, phase-amplitude coupling (PAC) across frequencies, and long-range phase synchrony. To estimate LFP power, spectral power was calculated using the mtspecgramc function in the Chronux toolbox and then converted to decibels (dB) for visualization. PAC strength was measured with an optimized version of the modulation index (MI) developed by Tort et al.[53]. To examine dynamic changes in neural synchrony, we adapted methods from Lachaux et al. to compute the phase-locking value (PLV) between brain regions[17,54]. For each trial, LFP signals were transformed into the time-frequency domain using the Morlet wavelet transform, and PLV was calculated with a sliding window (window size: 1 s, step size: 0.1 s). Within each window, PLV was calculated as follows:

$$PLV_{(f,t)} = \left| \frac{1}{n} \sum_{k=1}^{n} e^{i(P_{electrod1}(f,k) - P_{electrode2}(f,k))} \right| \quad (1)$$

where f is frequency, t is the center time point of the window, n is the number of samples within the window, k is the sample index, and $P_{electrod1}$ and $P_{electrod2}$ represent the instantaneous phases of the signals from the two electrodes at frequency f and time t, respectively. For optogenetic tagging, the same system was configured to record at a rate of 30 kHz.

Fiber photometry

Fiber photometry was conducted using a commercial Fiber photometry DAQ system (QAXK-FPS-LASER, Thinker Tech, China). A 488 nm laser, with a power of less than 45 μW at the fiber tip, was used. To prevent photobleaching, recordings were limited to 15 mi per mnouse per day. Signals were sampled at 100 Hz. To ensure consistency in data processing across training days, we analyzed only the signals from hit trials. The dataset, collected from day1 to day6 of training, included 200 trials from six mice. Data were processed offline with custom MATLAB scripts. For each trial, signals were baseline Z-scored to the 2 s pre-stimulus period. To quantify the relationship between γ-synchrony and VTA^{DA} neuron activity, fiber-photometry traces were downsampled to 10 Hz to match the temporal resolution of the PLV. 20–40 Hz PLV was averaged, and within the stimulus epoch, we extracted the maximum PLV and the maximum photometry amplitude. Daily mean signal amplitudes were then used to assess the linear correlation between PLV and photometry activity. To compare response onset latencies of γ-synchrony and VTA^{DA} neuron responses to external stimuli, the response onset index for each trial was computed using MATLAB's built-in function gradient, and the latency difference between the two signals was defined as Lag = Time $_{\text{max gradient PLV}}$ - Time $_{\text{max gradient VTA}}$.

Optogenetic manipulation. A 470 nm laser (Lasever, China) was used for optogenetic activation, and a 589 nm laser (Lasever, China) was used for optogenetic inhibition. The timing of laser application was precisely controlled by a custom-made, Arduino-based circuit. For activating VTA^{DA} neurons, a 470 nm laser (7 mW/mm² at the fiber tip, 20 Hz pulse train, 10 ms per pulse) was applied. For optotagging LEC^{5} neurons, a 470 nm laser (5 mW/mm² at the fiber tip, 20 Hz pulse train, 10 ms per pulse) was used. To inhibit VTA^{DA}-LEC^{5} axons during the CS-US associative learning task, a constant 589 nm laser (7 mW/mm² at the fiber tip) was applied in each trial for 3 s, covering the delay period and response window. For simultaneous inhibition of VTA^{DA}-LEC^{5} axons and stimulation of LEC^{5} neurons, 589 nm (7 mW/mm² at the fiber tip, constant) and 473 nm (7 mW/mm² at the fiber tip, 30 Hz pulse train, 10 ms per pulse) lasers were combined using a 2 × 2, 50:50 multi-mode fiber coupler (TT200FL2A, Thorlabs, USA). These hybrid lasers were applied together for 3 s, covering the delay period and response window. For the CPP experiment, a 473 nm laser (7 mW/mm² at the fiber tip, 30 Hz pulse train, 10 ms per pulse) was used for 10 min to activate LEC^{5} neurons. For the CFC experiment, the same 473 nm laser (7 mW/mm² at the fiber tip, 30 Hz pulse train, 10 ms per pulse) was used for 3 min to activate LEC^{5} neurons.

Optotagging and spike sorting

All optotagging and spike recordings were performed with an Apollo system (v1.0, Bio-Signal Technologies, USA). The raw signals were processed through a preamplifier to distinguish spike activities (using a 300 Hz high-pass filter, sampled at 30 kHz) and LFPs (using a 200 Hz low-pass filter, sampled at 1 kHz). Before each recording session, 20 Hz blue laser trains lasting 1 s were delivered to identify oChIEF-expressing units. A unit was classified as an Rbp4-positive neuron if it showed high reliability of laser-evoked spikes with a short latency (<10 ms). We further confirmed their identity by comparing waveform correlations between laser-evoked and spontaneous spikes. A total of 11 Rbp4-positive neurons were isolated from six mice, with most recorded from a single tetrode. Spike sorting was done manually using Plexon Offline Sorter (version 4.5.0) and MATLAB scripts from https://github.com/MathWorks-Teaching-Resources/Electrophysiology-Tutorial-for-Neuroscience/releases/tag/v1.2. Mean spike activity was averaged in 100-ms bins. The circular vector length, used to measure the degree of spike-oscillation phase locking, was calculated using a MATLAB toolbox for circular statistics[55].

2P calcium imaging

During the CS-US associative learning task, calcium imaging was performed in awake, head-fixed mice using a two-photon microscope (25× water-immersion objective, NA 1.05, FVMPE-RS, Olympus, Japan). The excitation wavelength was 960 nm (Ti: sapphire, approximately 10 mW at the brain, held constant). Emission was collected at 380–480, 500–540, and 560–650 nm. The field of view measured 509 × 509 µm. Imaging was captured at 30 frames per second (fps).

For 2P imaging of RSC neurons, to minimize GCaMP6f photobleaching artifacts, imaging sessions were conducted only on day1 (10 trials each for CS-only and US-only), day3 (30 CS-US trials), and day7 (30 CS-US trials). On the remaining days, the mice underwent only 100 CS-US training trials without 2P imaging.

In accordance with the aforementioned imaging schedule, an additional two days of imaging were conducted on a group of Rbp4-Cre mice in the chemogenetic inhibition experiment depicted in Supplementary Fig. 9. These imaging sessions were performed one hour after the injection of clozapine N-oxide (CNO) on day8 and the injection of phosphate-buffered saline (PBS) on day9. The bilateral inhibition of the LEC[5] was achieved via the intraperitoneal (i.p.) administration of CNO (0.6 mg/ml, 2 mg/kg; dissolved in PBS, APExBIO, USA).

To ensure consistency between images taken on different days, we applied both translational and rotational corrections to account for brain movement and variations in head-fixation angles[56]. For the extraction of fluorescence signals from individual neurons, GCaMP6f-positive cells were manually annotated using ImageJ.

For 2P imaging of LEC[5]-RSC axons, due to the low photostability of untargeted axonal GCaMP6s and the difficulty of maintaining the same field of view across days, the response window was replaced with direct reward delivery (exemption from triggering) to speed up the imaging process. Mice first completed ten trials each of CS-only and US-only stimuli, followed by a training session consisting of a batch of 15 association trials with one drop of water (1st Asso.), five double US trials with two drops of water (D-US), and five US omission trials without water delivery (US-omi.). Then, a second batch of five association trials with one drop of water (2nd Asso.) was conducted. The noise suppression DeepCAD-RT approach was used for both RSC neurons and LEC[5]-RSC axons (https://github.com/cabooster/DeepCAD-RT)[57].

Conditioned place preference (CPP)

Mice were kept on a food-restricted diet throughout the entire experiment to maintain a body weight of 80–85%. Training started when the body weight reached 85%. The mice were monitored daily for general health. Behavioral testing was conducted in a custom two-chamber box (60 × 40 cm) with a removable divider that allowed for either connection or separation of the compartments. One compartment had black interior walls (the dark box), while the other had white interior walls (the light box). On the first day, mice were gently placed at the junction between the two compartments and allowed to explore both sides for 20 min. Baseline exploration showed a spontaneous preference for the dark compartment. On the second day, the compartments were separated with the divider. Mice first experienced a 10-min exploration in the dark box without buried food or optogenetic stimulation, followed by a 10-min foraging period in the bright box containing buried food. During foraging, bilateral LEC[5] neurons were stimulated with a 473 nm laser pulsed at 30 Hz (other optical parameters as specified in the Optogenetic manipulation section). During the recall tests on day3 and day5, with the divider removed, mice were gently placed at the compartment boundary and allowed to explore both sides for 20 min without buried food and without optogenetic stimulation. The time spent in each compartment was measured to assess the preference index. A ceiling-mounted camera (Logitech) recorded the mice's trajectories with high spatial and temporal resolution (30 frames per second). Behavioral trajectories were analyzed offline from video recordings using custom MATLAB scripts, and the duration of occupancy within each compartment was quantified. During the CPP task, a preference index was calculated for each mouse as $(Time_{light} - Time_{dark}) / (Time_{light} + Time_{dark})$ and summarized at the group level.

Contextual Fear Conditioning (CFC)

Experiments were carried out using the Multi Conditioning System (version 2.0, TSE Systems, Germany). During the conditioning session, mice were placed in the conditioning chamber (20 × 20 cm) and allowed to habituate for 3 min before receiving a mild foot shock (0.3 mA for 2 s). Immediately afterward, LEC[5] neurons were subjected to optogenetic stimulation with a 470 nm laser pulsed at 30 Hz for 3 min (other optical parameters as specified in the Optogenetics section). Freezing behavior was measured during two recall tests, each lasting 3 min, conducted immediately after conditioning and 24 h later. For the CFC task, animal behaviors were automatically recorded by the TSE Multi Conditioning System. The freezing level was assessed using the system's built-in analysis software, which employed the manufacturer's default freezing-detection algorithm with a minimum bout duration of 1 s. The results were then exported for further analysis. For each mouse, the freezing percentage was calculated as the proportion of time classified as freezing during the recall test.

Mandarin phrase-reading task

Each patient undergoing awake glioma resection participated in a two-character Mandarin phrase-reading task. Each trial had three sequential phases: an initial 1 s fixation period with a central crosshair displayed, followed by a 2 s perception phase during which Mandarin lexical stimuli appeared in green, and finally a 1.5 s production phase. At the start of the production phase, the stimulus color changed from green to red, prompting participants to vocalize the displayed text. The stimulation included four real phrases and four pseudo-phrases. Each phrase was presented in sessions of three repetitions, with each session consisting of two blocks and a few-minute rest between sessions, resulting in six total repetitions per phrase.

ECoG signals were recorded at a sampling rate of 3052 Hz using Tucker-Davis Technologies (TDT) multichannel amplifiers integrated with digital acquisition systems. The timing of visual stimuli was precisely synchronized using photodiode monitoring of the presentation screen, and this information was directly fed into the TDT circuitry to ensure accurate alignment with the ECoG data. Online referencing was performed with a skull electrode placed at the Cz location, and no additional re-referencing was done during acquisition. Due to variability in electrode placement across patients, all electrode channels were first classified into three groups based on their anatomical location: frontal, temporal, and parietal. Before proceeding with further analysis, channels with issues were identified and excluded from the dataset. To handle the high dimensionality and computational demands of the PLV analysis, a subset of 30 channels per patient was selected (Supplementary Fig. 8 and Supplementary Table 4). Specifically, we aimed for a balanced allocation of 10 channels per lobe; however, if a lobe contained fewer than 10 valid channels, the unfulfilled quota was reallocated to the region with the highest number of available channels. To ensure global representativeness, we implemented a greedy farthest-point sampling strategy within each region to maximize spatial coverage uniformity. Operationally, the 256 channels were mapped onto a 16×16 planar array; the first channel was selected based on minimal Euclidean distance to the geometric center, and subsequent channels were iteratively chosen to maximize the minimum Euclidean distance to the set of already selected channels. Raw data from these channels were downsampled to 1 kHz, and power-line noise was filtered out. Time-frequency

analysis was then conducted using the Chronux toolbox (version 2.12), and PLV was calculated for each trial according to Eq. (1). For the statistical analysis, we specifically selected electrode pairs that exhibited at least a 5% increase in PLV during pseudo-phrase presentation in the first block. PLV values from all trials were first averaged for each experimental block and for each phrase, respectively. This averaging was performed within the 20–25 Hz frequency band and the 0.5 s window following phrase onset. The mean PLV values were then compared using a paired one-way ANOVA, followed by post-hoc tests for multiple comparisons.

Histology

Mice were anesthetized with 3% tribromoethanol and perfused with PBS, followed by 4% paraformaldehyde (PFA, 158127, Sigma-Aldrich) in PBS. Brain tissue was post-fixed overnight. Coronal brain sections (50 μm thick) were cut using a Leica vibratome (VT1000 S). For staining, sections were first washed three times with PBS and then blocked in a solution of 10% goat serum (EE0008, SparkJade, China) in PBS for 30 min at room temperature. Next, the sections were incubated overnight at 4 °C with a primary antibody. After three washes, the sections were incubated for 1 h at room temperature with a secondary antibody. The antibodies used in the experiment are listed in Table S2. All antibodies were used at a dilution of 1:1000. After three additional washes, the sections were coverslipped using Fluoromount-G with DAPI (36308ES20, Yeasen, China). All samples were imaged on a spinning disk confocal microscope (Nikon CSU-W1, Nikon, Japan) using a 20×/0.75 objective. The animals used in the recording experiments were sacrificed to verify the position of the implants or the areas of virus infection.

Statistical analyses

Data are shown as mean ± SEM. Statistical analyses were performed in GraphPad Prism (v11.0.0, GraphPad Software) and MATLAB (2020b). Two-tailed paired or unpaired *t*-tests, multiple *t*-tests, one-way ANOVA, two-way ANOVA, and the Kolmogorov–Smirnov test were used as appropriate. The significance threshold was set at $P < 0.05$. Significance is denoted as follows: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$. All statistical results are listed in the Supplementary Table 1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Datasets used in this study are available in the Figshare database. https://doi.org/10.6084/m9.figshare.30372145 Source data are provided with this paper.

Code availability

Custom MATLAB codes supporting this study are available in the Figshare database. https://doi.org/10.6084/m9.figshare.30372145.

References

1. Sutton, R. S. & Barto, A. G. Toward a modern theory of adaptive networks: expectation and prediction. *Psychol. Rev.* **88**, 135–170 (1981).
2. Schultz, W., Dayan, P. & Montague, P. R. A neural substrate of prediction and reward. *Science* **275**, 1593–1599 (1997).
3. Montague, P. R., Dayan, P. & Sejnowski, T. J. A framework for mesencephalic dopamine systems based on predictive Hebbian learning. *J. Neurosci.* **16**, 1936–1947 (1996).
4. Glimcher, P. W. Understanding dopamine and reinforcement learning: the dopamine reward prediction error hypothesis. *Proc. Natl. Acad. Sci. USA* **108**, 15647–15654 (2011).
5. Liu, X. et al. Optogenetic stimulation of a hippocampal engram activates fear memory recall. *Nature* **484**, 381–385 (2012).
6. Josselyn, S. A. & Tonegawa, S. Memory engrams: Recalling the past and imagining the future. *Science* **367** https://doi.org/10.1126/science.aaw4325 (2020).
7. Goode, T. D., Tanaka, K. Z., Sahay, A. & McHugh, T. J. An integrated index: engrams, place cells, and hippocampal memory. *Neuron* **107**, 805–820 (2020).
8. Roy, D. S. et al. Brain-wide mapping reveals that engrams for a single memory are distributed across multiple brain regions. *Nat. Commun.* **13**, 1799 (2022).
9. Xie, H. et al. In vivo imaging of immediate early gene expression reveals layer-specific memory traces in the mammalian brain. *Proc. Natl. Acad. Sci. USA* **111**, 2788–2793 (2014).
10. Buzsáki, G. & Vöröslakos, M. Brain rhythms have come of age. *Neuron* **111**, 922–926 (2023).
11. Fries, P. A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends Cogn. Sci.* **9**, 474–480 (2005).
12. Fries, P. Rhythms for cognition: communication through coherence. *Neuron* **88**, 220–235 (2015).
13. Palmigiano, A., Geisel, T., Wolf, F. & Battaglia, D. Flexible information routing by transient synchrony. *Nat. Neurosci.* **20**, 1014–1022 (2017).
14. Sohal, V. S. How close are we to understanding what (if anything) γ oscillations do in cortical circuits? *J. Neurosci.* **36**, 10489–10495 (2016).
15. Bastos, A. M. et al. Canonical microcircuits for predictive coding. *Neuron* **76**, 695–711 (2012).
16. Grover, S., Wen, W., Viswanathan, V., Gill, C. T. & Reinhart, R. M. G. Long-lasting, dissociable improvements in working memory and long-term memory in older adults with repetitive neuromodulation. *Nat. Neurosci.* **25**, 1237–1246 (2022).
17. Luo, W. et al. Acquiring new memories in neocortex of hippocampal-lesioned mice. *Nat. Commun.* **13**, 1601 (2022).
18. Hamid, A. A. et al. Mesolimbic dopamine signals the value of work. *Nat. Neurosci.* **19**, 117–126 (2016).
19. Chang, C. Y. et al. Brief optogenetic inhibition of dopamine neurons mimics endogenous negative reward prediction errors. *Nat. Neurosci.* **19**, 111–116 (2016).
20. Sharpe, M. J. et al. Dopamine transients are sufficient and necessary for acquisition of model-based associations. *Nat. Neurosci.* **20**, 735–742 (2017).
21. Saunders, B. T., Richard, J. M., Margolis, E. B. & Janak, P. H. Dopamine neurons create Pavlovian conditioned stimuli with circuit-defined motivational properties. *Nat. Neurosci.* **21**, 1072–1083 (2018).
22. Lee, K. et al. Temporally restricted dopaminergic control of reward-conditioned movements. *Nat. Neurosci.* **23**, 209–216 (2020).
23. Morrens, J., Aydin, C., Janse van Rensburg, A., Esquivelzeta Rabell, J. & Haesler, S. Cue-evoked dopamine promotes conditioned responding during learning. *Neuron* **106**, 142–153 (2020).
24. Lak, A. et al. Dopaminergic and prefrontal basis of learning from sensory confidence and reward value. *Neuron* **105**, 700–711.e706 (2020).
25. Stauffer, W. R. et al. Dopamine neuron-specific optogenetic stimulation in rhesus macaques. *Cell* **166**, 1564–1571 (2016).
26. Luan, L. et al. Ultraflexible nanoelectronic probes form reliable, glial scar-free neural integration. *Sci. Adv.* **3**, e1601966 (2017).
27. Bennett, J. E. M., Philippides, A. & Nowotny, T. Learning with reinforcement prediction errors in a model of the Drosophila mushroom body. *Nat. Commun.* **12**, 2569 (2021).
28. Gadagkar, V. et al. Dopamine neurons encode performance error in singing birds. *Science* **354**, 1278–1282 (2016).

29. Jang, A. I., Nassar, M. R., Dillon, D. G. & Frank, M. J. Positive reward prediction errors during decision-making strengthen memory encoding. *Nat. Hum. Behav.* **3**, 719–732 (2019).
30. Solié, C., Girard, B., Righetti, B., Tapparel, M. & Bellone, C. VTA dopamine neuron activity encodes social interaction and promotes reinforcement learning through social prediction error. *Nat. Neurosci.* **25**, 86–97 (2022).
31. Xie, Y., Huang, L., Corona, A., Pagliaro, A. H. & Shea, S. D. A dopaminergic reward prediction error signal shapes maternal behavior in mice. *Neuron* **111**, 557–570.e557 (2023).
32. Akiti, K. et al. Striatal dopamine explains novelty-induced behavioral dynamics and individual variability in threat prediction. *Neuron* **110**, 3789–3804 e3789 (2022).
33. Konova, A. B. et al. Reduced neural encoding of utility prediction errors in cocaine addiction. *Neuron* **111**, 4058–4070.e4056 (2023).
34. Bott, F. S. et al. Local brain oscillations and interregional connectivity differentially serve sensory and expectation effects on pain. *Sci. Adv.* **9**, eadd7572 (2023).
35. Sohoglu, E., Beckers, L. & Davis, M. H. Convergent neural signatures of speech prediction error are a biological marker for spoken word recognition. *Nat. Commun.* **15**, 9984 (2024).
36. O'Toole, S. M., Oyibo, H. K. & Keller, G. B. Molecularly targetable cell types in mouse visual cortex have distinguishable prediction error responses. *Neuron* **111**, 2918–2928 (2023).
37. Furutachi, S., Franklin, A. D., Aldea, A. M., Mrsic-Flogel, T. D. & Hofer, S. B. Cooperative thalamocortical circuit mechanism for sensory prediction errors. *Nature* **633**, 398–406 (2024).
38. Drieu, C. et al. Rapid emergence of latent knowledge in the sensory cortex drives learning. *Nature* **641**, 960–970 (2025).
39. Kanter, B. R., Lykken, C. M., Polti, I., Moser, M. B. & Moser, E. I. Event structure sculpts neural population dynamics in the lateral entorhinal cortex. *Science* **388**, eadr0927 (2025).
40. Igarashi, K. M., Lu, L., Colgin, L. L., Moser, M. B. & Moser, E. I. Coordination of entorhinal-hippocampal ensemble activity during associative learning. *Nature* **510**, 143–147 (2014).
41. Issa, J. B., Radvansky, B. A., Xuan, F. & Dombeck, D. A. Lateral entorhinal cortex subpopulations represent experiential epochs surrounding reward. *Nat. Neurosci.* **27**, 536–546 (2024).
42. Lee, J. Y. et al. Dopamine facilitates associative memory encoding in the entorhinal cortex. *Nature* **598**, 321–326 (2021).
43. Jun, H. et al. Prefrontal and lateral entorhinal neurons co-dependently learn item-outcome rules. *Nature* **633**, 864–871 (2024).
44. Igarashi, K. M. Entorhinal cortex dysfunction in Alzheimer's disease. *Trends Neurosci.* **46**, 124–136 (2023).
45. Tang, J. C. Y. et al. Dynamic behaviour restructuring mediates dopamine-dependent credit assignment. *Nature* **626**, 583–592 (2024).
46. Rouhani, N. & Niv, Y. Signed and unsigned reward prediction errors dynamically enhance learning and memory. *eLife* **10**, https://doi.org/10.7554/eLife.61077 (2021).
47. Fernandez-Ruiz, A., Sirota, A., Lopes-dos-Santos, V. & Dupret, D. Over and above frequency: Gamma oscillations as units of neural circuit operations. *Neuron* **111**, 936–953 (2023).
48. Luo, W. & Guan, J.-S. Do brain oscillations orchestrate memory? *Brain Sci. Adv.* **4**, 16–33 (2018).
49. Lee, S. Y. et al. Cell-class-specific electric field entrainment of neural activity. *Neuron* **112**, 2614–2630.e2615 (2024).
50. Sohal, V. S., Zhang, F., Yizhar, O. & Deisseroth, K. Parvalbumin neurons and gamma rhythms enhance cortical circuit performance. *Nature* **459**, 698–702 (2009).
51. Veit, J., Handy, G., Mossing, D. P., Doiron, B. & Adesnik, H. Cortical VIP neurons locally control the gain but globally control the coherence of gamma band rhythms. *Neuron* **111**, 405–417 (2023).
52. Yin, R. et al. Chronic co-implantation of ultraflexible neural electrodes and a cranial window. *Neurophotonics* **9**, 032204 (2022).
53. Tort, A. B., Komorowski, R., Eichenbaum, H. & Kopell, N. Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *J. Neurophysiol.* **104**, 1195–1210 (2010).
54. Lachaux, J. P., Rodriguez, E., Martinerie, J. & Varela, F. J. Measuring phase synchrony in brain signals. *Hum. Brain Mapp.* **8**, 194–208 (1999).
55. Berens, P. CircStat: a MATLAB toolbox for circular statistics. *J. Stat. Softw.* **31**, 1–21 (2009).
56. Liu, C. et al. Hippocampus alters visual representation to encode new memory. *Cell Rep.* **44**, 115594 (2025).
57. Li, X. et al. Real-time denoising enables high-sensitivity fluorescence time-lapse imaging beyond the shot-noise limit. *Nat. Biotechnol.* **41**, 282–292 (2023).

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Author contributions

D.Y. and J.-S.G. conceived the project. D.Y. performed the experiments with input from Z.W. (fiber photometry and two-photon calcium imaging with simultaneous LFP recording) and S.Z. (anterograde tracing). Z.J.W., J.L. and Y.L. collected the human ECoG data. H.M. and F.H. provided the ultraflexible electrode arrays. D.Y. wrote the original manuscript with comments from other authors. H.X. contributes to the establishment of methodology and data analysis in 2P microscopy. H.X. and J.-S.G. supervised this project.

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Competing interests

The authors declare no competing interests.

Additional information

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